Computer Vision and Image Understanding 269 (2026) 104791

Contents lists available at ScienceDirect

Computer Vision and Image Understanding

journal homepage: www.elsevier.com/locate/cviu

Self-attention as the backbone: A survey on Vision Transformers

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ARTICLE INFO

Communicated by Paolo Rota

Keywords:
Vision Transformer
Attention mechanism
Image classification
Foundation models
State space models
Multimodal learning
Survey

ABSTRACT

The human cognitive system efficiently processes complex environments by focusing on salient regions, inspiring attention mechanisms that amplify critical information. Deep learning models integrate attention to enhance performance across tasks, with self-attention gaining prominence through its remarkable success in Transformer-based natural language processing. This success propelled its adoption in computer vision, where Transformer-based models surpass conventional deep networks on multiple benchmarks. Recently, large-scale pretraining has enabled vision foundation models, where self-attention acts as a scalable backbone for learning transferable visual representations across diverse tasks. By modeling global dependencies without heavy inductive biases, self-attention supports flexible and expressive representation learning. However, Vision Transformers (ViTs) rely on large parameter counts and high computational budgets, and their quadratic complexity with respect to input size restricts scalability and real-world deployment. Consequently, substantial research has focused on reducing computational overhead while preserving or improving performance through self-attention optimization. In parallel, State Space Models (SSMs) have emerged as an alternative paradigm, replacing attention with linear-time recurrence for efficient long-range modeling. This survey provides a comprehensive review of advanced backbone techniques that improve ViTs through novel self-attention designs or hybrid integrations, mainly for image classification, including their transfer to multimodal large language models, generative frameworks, and emerging SSM-based architectures, alongside ViT explainability approaches. We first outline self-attention and Transformers, then introduce a taxonomy of self-attention variants. Finally, we systematically compare attention architectures in terms of design, accuracy, and computational efficiency, and discuss current limitations and future research directions.

1. Introduction

Attention is recognized in cognitive science as a fundamental mechanism that allows the human brain to selectively process relevant information while filtering out distractions (Rensink, 2000; Corbetta and Shulman, 2002). Although the brain is exposed to an overwhelming amount of sensory data from its surroundings, it does not process everything simultaneously. Instead, it prioritizes the information dynamically (Grootswagers et al., 2023). Inspired by these selective principles, researchers sought to develop attention mechanisms for machines. These mechanisms were initially used in isolation and later incorporated into deep learning models such as the Convolutional Neural Network (CNN) to achieve better performance. The concept of such machine attention was first proposed by Bahdanau et al. (2015), which plays a crucial role in the recent history of deep learning. Machine attention operates by dynamically adjusting feature weights, prioritizing them based on the relevance of the given input. Recent advancements have shown significant performance improvements in many artificial intelligence fields such as Natural Language Processing (NLP) (Touvron et al., 2023), vision (Chen et al., 2023a) and time series forecasting (Zhou et al., 2021c).

Before attention mechanisms gained popularity, CNNs and Long-Short-Term Memory (LSTM) models were dominant in computer vision and Natural Language Processing (NLP), respectively. In NLP tasks, the bottleneck problem in encoder–decoder architectures arises when the entire input sequence is compressed into a single fixed-length vector, limiting the decoder's ability to retain and utilize important information, especially for long sequences. Bahdanau et al. (2015) address this bottleneck problem by allowing the decoder to dynamically focus on the most relevant parts of the encoded input using attention. In vision, CNNs dominated due to their ability to capture hierarchies in data. However, CNNs used fixed-sized kernels to extract information locally, making it a challenge to capture global context effectively. Contrarily, the attention mechanism allows the models to capture more features in an efficient way globally. This mechanism effectively computes the importance of different parts of the input which trains the models

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https://doi.org/10.1016/j.cviu.2026.104791
Received 24 September 2025; Received in revised form 24 February 2026; Accepted 22 April 2026
Available online 29 April 2026
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more effectively. Therefore, the attention mechanisms are highly effective at modeling complex relationships between input features, often outperforming CNNs in tasks involving long-range dependencies

The real innovation came with the introduction of the Transformer model proposed by Vaswani et al. (2017) which revolutionized the artificial intelligence field. Transformer-based models (Devlin et al., 2019; Radford et al., 2018; Dehghani et al., 2019) demonstrated a great performance boost in NLP tasks i.e. text classification, problem-solving and machine translation. Their success in processing sequential data encouraged researchers to explore their potential in computer vision tasks. While early attempts to use images with Transformers (Parmar et al., 2018; Ho et al., 2019; Li et al., 2019) showed promising outcomes but most of them could not achieve state-of-the-art results. The breakthrough in computer vision occurred when Dosovitskiy et al. (2021) proposed a Vision Transformer (ViT) which directly inherited the original Transformer architecture (Vaswani et al., 2017) and applied it to image classification. Through a combination of novel design and huge amounts of training data, it outperformed state-of-the-art Convolutional Neural Networks (CNNs), proving that the self-attention mechanism could be effectively used for computer vision tasks. Consequently, this mechanism is now at the forefront of artificial intelligence research, helping to build new versatile, scalable and efficient models.

The core innovation of the Transformer lies in the self-attention technique which reshaped the field of artificial intelligence. Despite its impact, this method has certain limitations and flaws. To mitigate these challenges, numerous studies have been published in recent years to improve self-attention. Since the release of ViT, new publications on vision Transformers (VTs)[1] have been increasing exponentially. This rapid expansion emphasizes the need for an updated and comprehensive survey that includes the most recent advancements in the field, not covered in previous works (Khan et al., 2022; Han et al., 2022).

Previous major surveys (Han et al., 2022), Liu et al. (2023c) have provided a broad overview of VTs while categorizing the models based on vision tasks, the present work is more narrowly focused on image classification backbones, their transferability to multimodal large language models and generative pipelines and post-vit architectures, to keep the discussion concentrated on self-attention in ViT as the foundational architectural backbone. The focus is important as many Transformer-based architectures for other vision tasks draw inspiration directly from such backbone framework (Wang et al., 2021; Liu et al., 2021). This deliberate scope allows us to cover a larger number of relevant publications compared to other surveys in the classification task while also adding very recent developments (Fan et al., 2025; Sun et al., 2025). Furthermore, a refined and more elaborated taxonomy is proposed in this survey, offering detailed and clearer classification criteria that not only reflect existing architectural variations but also introduce novel categories that have not been systematically addressed in previous surveys (Khan et al., 2022, 2023) which help in understanding architectural variations. For each category, we systematically outline the advantages and disadvantages, providing an analytical perspective that facilitates comparative evaluation. We also include detailed diagrams that show the internal mechanisms and structural differences between self-attention mechanisms. Additionally, we examine the various open challenges in depth and outlining more detailed future research directions compared to prior surveys.

This survey primarily concentrates on models that position themselves as the backbone in VTs, developed for image classification. This focus is motivated by the fact that image classification serves as the core task in computer vision, acting as the benchmark such as ResNet (He et al., 2016) and ViT (Dosovitskiy et al., 2021). These models are further evolved to improve image classification and other vision tasks such as object detection (Lin et al., 2017), semantic segmentation (Zhang et al., 2022b), or image super-resolution (Lu et al., 2022b), thanks to their modular and transferable design while allowing them to serve as a backbone for newer models (Wang et al., 2021). In addition to that, some notable works are also reviewed that introduce distinctive attention mechanisms built specifically for other tasks (Xie et al., 2021; Wang et al., 2022d). Recent advances increasingly emphasize large-scale pretraining on massive datasets, giving rise to vision foundation models. Models such as DINOv2 (Oquab et al., 2024), SigLIP (Zhai et al., 2023), and AIM (El-Nouby et al., 2024) demonstrate that self-attention, when combined with data-centric scaling, can produce highly transferable visual representations. This shift highlights that the architectural innovations in self-attention remain crucial, but their impact is often realized within large-scale training regimes.

Furthermore, this survey mainly focuses on novel research techniques that specifically enhance self-attention mechanisms, as self-attention forms the core of many state-of-the-art models and has become a central focus of recent architectural innovations and research efforts. Our research shows that the most recent literature on VTs predominantly concentrates on modifying or enhancing self-attention mechanisms, rather than introducing entirely new forms of attention. The distinction here is important: instead of proposing new types of attention i.e. Guo et al. (2022b) Guo et al. (2022c), which use different architectures than self-attention, most contributions focus on refining the existing self-attention framework based on ViT. Consequently, this survey centers on techniques that alter self-attention, reflecting the main trajectory of major VTs research. The primary aim is to provide a comprehensive overview of recent and influential works that focus on improving the performance and computational efficiency of VTs. By consolidating these advancements, this survey offers valuable insights that can inform and guide future research directions in the design and optimization of Transformer-based architectures.

While this survey focuses on innovations within the self-attention framework, we also acknowledge the growing shift toward State Space Models (SSMs). Architectures such as Vim (Zhu et al., 2024a) and VMamba (Liu et al., 2024b) achieve global receptive fields with linear complexity, offering attention-free alternatives. Many techniques taxonomized in this survey are now being integrated into hybrid SSM–Transformer backbones. In addition, we incorporate recent works on interpretability and explainability in VTs which provide a systematic tools to analyze and evaluate the faithfulness and robustness of attention mechanisms.

To ensure comprehensive coverage of recent advances in self-attention for VTs, we adopted a structured literature selection strategy. We reviewed papers published between 2020 and early 2026 in leading computer vision and machine learning venues, including CVPR, ICCV, NeurIPS, TPAMI, IJCV, and related high-impact conferences and journals. Keyword-based searches (e.g., “Vision Transformer”, “ViT”, “self-attention”, “attention”, “tokens” and many more) were conducted across conference proceedings, Google Scholar, and arXiv. We also identified influential works through citation analysis, including backward and forward tracking of highly cited papers. We included studies that proposed architectural modifications or reformulations of self-attention for vision tasks and provided empirical validation on standard benchmark of ImageNet (Deng et al., 2009). Eligible works were peer-reviewed papers or highly cited preprints with clear community impact. We excluded studies that applied self-attention to niche applications without modifying the underlying mechanism, focused solely on NLP, lacked experimental validation, or offered only incremental improvements without clear gains in accuracy, efficiency, or complexity.

Vision Transformer models are classified by analyzing the fundamental features of self-attention. In this survey, we discuss the evolution of VTs by dividing them into multiple categories based on architectural modifications and approaches for efficiency and performance improvements.

[1] To clarify terminology, we refer to the original Vision Transformer as ViT. When describing the general category of vision-based Transformers we use the acronym VTs.

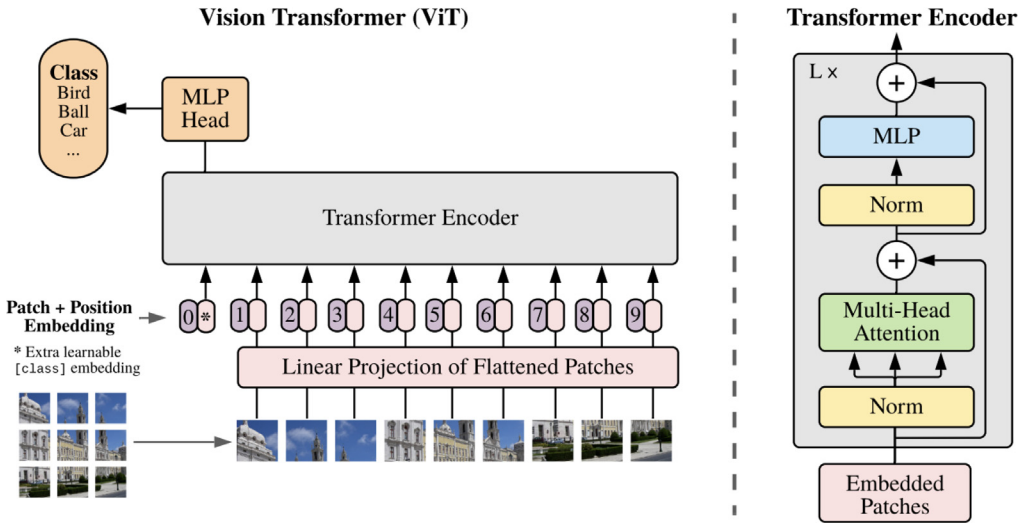


Fig. 1. Illustration of the original Vision Transformer (ViT) architecture taken from the original paper (Dosovitskiy et al., 2021). Left: The input image is divided into fixed-size patches, which are linearly embedded, combined with positional embeddings and a class token, and transferred to a standard Transformer encoder. Right: The internal structure of the Transformer encoder, consisting of multi-head self-attention, normalization and a feed-forward MLP block.

The article is structured as follows: We begin with Section 2 where the fundamental concepts underlying the Transformer are discussed and then are further elaborated with reference to the ViT. Next, we review all the major techniques and advancements for VTs in Section 3. Next in Section 4 we consider self-attention in vision foundation models. We then discuss the Post-ViT age and SSMs in Section 5. We broaden the task scope in Section 6. Next, we consider interpretability and explainability in Section 7. We then compare the performance and efficiency of these models in Section 8. Challenges and future directions are covered in Section 9. Finally, we conclude in Section 10.

2. The foundation of the transformer

As discussed in Section 1, attention as a technique has seen a progression from independent use, to CNNs and RNNs, to its most recent incarnation in the Transformer, and subsequent VTs models. In this section, we elaborate on the use of attention with reference to ViT, and then consider multi-head self-attention.

2.1. Self-attention

After the impact of the attention mechanism on deep learning, researchers tried to improve the attention for different applications. Among these improvements (Cheng et al., 2016) proposed a new technique called self-attention for natural language processing tasks which allowed a model to calculate the relevance between different parts of the input. This mechanism was improved and used in the original Transformer (Vaswani et al., 2017). Unlike traditional attention mechanisms, which focus on external sources (e.g., encoder–decoder attention in seq2seq models), self-attention calculates attention scores within the same input sequence. The self-attention takes an input, divides it into multiple tokens and computes the interaction between all tokens of the input. These tokens are used to generate different vectors, named query, key, and value, with the same dimension. After being generated from different inputs, the vectors are arranged into three matrices, namely Q, K, and V. Then a similarity score is calculated using a comparison between a query matrix of one token and the key matrix of all tokens which determines how much attention each token should get. The value matrix is then weighted by its attention score and aggregated to produce an output for one token. Self-attention can be described as,

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{1}$$

where Q, K and V are the query, key, and value vectors, respectively, and d_k is the dimension of the key vector. The dot product calculates the attention score between two tokens. The softmax function normalizes these scores, converting them into attention weights. The pairwise comparison of each and every token captures a measure of similarity between all tokens. The architecture of self-attention can be seen in Fig. 4 (first from left). This technique allows the model to choose relevant tokens in a sequence without any constraint, making them highly flexible and adaptive to different tasks. Unlike traditional models that process information in a fixed order, self-attention allows each token to attend to every other token, helping to capture global dependencies. This ability makes self-attention useful for handling complex and long-sequence data.

2.1.1. Self-attention in vision transformer (ViT)

The Self-attention mechanism acts as the foundation of all Transformer models. After its success in natural language processing, it was adopted by Dosovitskiy et al. (2021) to operate on visual data with the Vision Transformer (ViT). In ViT, an input image is split into non-overlapping patches, analogous to the tokens in NLP. These are then flattened and projected into high-dimension embedding space. In ViT, similar to a vanilla Transformer, patches are projected into query, key and value vectors and the same equation is applied for the similarity score. Additionally, positional encodings are added to patch embeddings to keep the relative position of patches intact. CNNs operate using convolution filters to process local neighborhoods while self-attention in ViT helps the model to capture global features of an image which allows the model to focus more on complex regions of the input, which gives the model a better understanding of an image.

2.2. Multi-head self-attention

Multi-head self-attention (MHSA) is another critical part of the ViT that improves the model's potential to capture more information from the image. The complete architecture of ViT can be seen in Fig. 1. In simple self-attention, each patch is compared with all other patches. However, MHSA extends this ability and uses multiple self-attention heads to work in parallel, as shown in Fig. 1 (right), which expands the model's understanding of the image. Each attention head computes its own attention score and then the outputs from all the heads are concatenated and linearly transformed as the final representation of the input image. It facilitates the capture of fine-grained details such

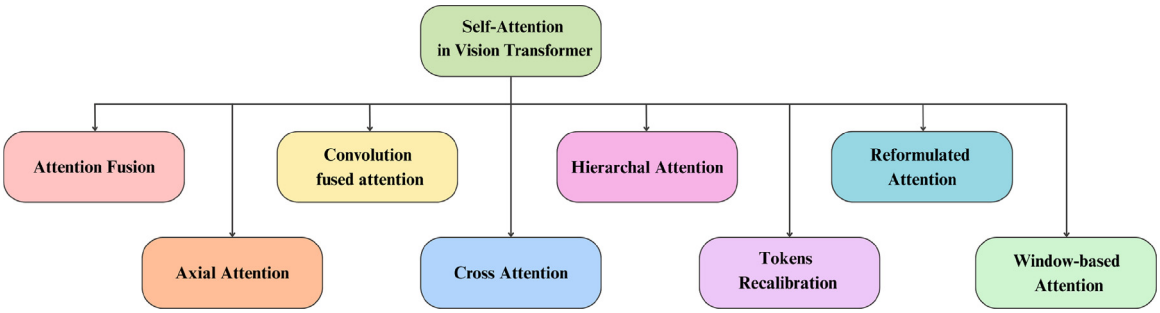


Fig. 2. Taxonomy of Self-Attention Techniques: the various components or mechanisms designed to enhance or modify the self-attention block. The existing approaches build upon self-attention by incorporating convolutional operations, window-based attention strategies, fusion and cross attention mechanisms, as well as modifications to tokens and the attention formula itself.

as textures, objects, shapes or edges, all within a single layer. This capability makes the model highly effective in different tasks like image classification, detection and segmentation.

Despite MHSA's state-of-the-art performance, it increases the computational complexity of ViT. To handle high-resolution images, the model needs to increase the number of patches and, as self attention is an operation with quadratic complexity, this can be challenging to operate efficiently. MHSA having parallel heads further amplifies this cost. As a result, training models with high-resolution images needs a significant amount of computational resources and memory. To tackle this issue as well as to improve the performance further, researchers have been trying to improve the attention mechanisms. Moreover, vanilla self-attention uses fixed-sized patches which is not applicable for many vision tasks and can be detrimental to performance. Therefore, researchers have been proposing new techniques to optimize the models and make them more scalable for a wide range of applications with improved performance.

3. Advanced vision transformers (VTs)

After the breakthrough performance of the ViT, a diverse range of attention-focused VTs were developed to extend or refine the ViT with the goal of improving performance and/or reducing the compute burden. Models exhibit diverse approaches to attention, with some specifically designed to enhance one aspect while others prioritize multiple. We categorize these models into a number of categories based on architectural modifications, attention strategies, and structural enhancements. In this review, we mainly concentrate on studies that only alter self-attention or introduce new componentry to improve accuracy, reduce computational complexity and memory consumption. It is extremely difficult to group these techniques under a single banner, as many employ hybrid approaches e.g., convolutions with cross-attention.

VTs can be broadly categorized based on scale. The vanilla ViT adopted a uniform scale input where the number of patches and feature dimensions were fixed throughout the architecture. This design limits the model in capturing finer details across multiple scales, and the arbitrary division of the image potentially results in singular semantic concepts being divided among tokens. Following the standard ViT, only a handful of studies continued improving VTs using uniform-scale structure, while the majority of the research shifted toward multi-scale or pyramid designs, and used variable-sized feature maps to make attention more efficient. This strategy gradually reduced the number of tokens while increasing the feature dimension, similarly to the convolution-pooling operation in most CNNs. These multi-scale VTs do not solely rely on a pyramid structure to improve performance but extend beyond it, combining other architectural innovations with multi-scaling in their models, such as Wu et al. (2022b), Hatamizadeh et al. (2024a). Given that incorporation, it is challenging to classify models strictly by scale. Consequently, this survey will discuss uniform and multi-scale approaches as they appear within different contexts.

The primary emphasis of the ViT was mainly on the long-range dependencies while overlooking the local feature extraction. This limitation was quickly identified and addressed through various techniques aimed at enhancing local context awareness, capturing fine-grained details within an input, while still maintaining the global context. Although some works continued to focus solely on the global context (Zhou et al., 2021a; Touvron et al., 2021b), recent studies have increasingly prioritized approaches that integrate both global and local perspectives. These approaches aim to leverage the strengths of both perspectives to achieve a more comprehensive understanding of spatial relationships across the visual data.

This survey categorizes models according to their primary self-attention modification, defined as the main architectural contribution of each work. Although some approaches integrate multiple design elements, each model is assigned to the category corresponding to its dominant attention innovation (e.g., convolutional integration, hierarchical abstraction, or token manipulation). The resulting taxonomy is mechanism-centered rather than name or application-based. Fig. 2 illustrates the defined attention components, and Table 1 lists the models included in this review. All surveyed models incorporate one or more of the components shown in Fig. 2, and the following sections group them by their ***Core Component***, i.e., the attention modification contributing most significantly to the novelty of the work.

One approach is to use convolution in tandem with self-attention. This is discussed next in Section 3.1. Following this, hierarchical attention mechanisms are explored. These methods introduce multiple levels of feature abstraction which are presented in Section 3.2. In Section 3.3, we shift focus to cross-attention strategies which enables interaction between distinct feature spaces. Window-based attention, where attention computation is restricted to local windows, is discussed in Section 3.4. Another important development is the use of axial attention, where attention is decomposed along different axes. This approach is detailed in Section 3.5. Attention mechanisms based on token modification are introduced in Section 3.6 where methods dynamically adjust the input tokens either by pruning, merging, or reweighting tokens. The review then moves to reformulations of self-attention, discussing parameterization of the original self-attention formula in Section 3.7. Finally, Section 3.8 addresses multiple types of attention integration approaches.

3.1. Self-attention fused with convolution

Despite the successful applications of VTs in various vision tasks, performance gaps persist between VTs and CNNs. This is largely due to the lower capacity of ViT to gather local context effectively. CNNs are excellent at capturing low-level information. Therefore, several ViT variants have been proposed to enhance locality by incorporating convolutional operations into attention-based architectures. This

Table 1
Categorization of Vision Transformer models according to their attention mechanisms. * denotes the *Core Category* of each model, while 'X' indicates secondary category affiliations. The table presents the top 10 highest-accuracy models within each model-size group.

Size	Method	Axial-Attn	Conv+Attn	Cross-Attn	Reform.	Hierarchical	Mixed-Attn	Tokens-Modf.	Windows
Extra-tiny	CoaT (Xu et al., 2021a)		*			X			
	LeViT (Graham et al., 2021)		X		*	X			
	EffiformerV2 (Li et al., 2023e)		*		X	X			
	Castling-ViT (You et al., 2023)				*		X		
	RVT (Mao et al., 2022)		X		*				
	CMT (Guo et al., 2022a)		*		X	X			
	EMO (Zhang et al., 2023)		X			X			*
	MixFormer (Chen et al., 2022b)		X			X			*
	LightViT (Huang et al., 2022)					X	*		X
	HRFormer (Yuan et al., 2021d)		X			X			*
Tiny	Pale-ViT (Wu et al., 2022a)	*				X			
	Inception (Si et al., 2022)		*		X	X			
	T2T-ViT (Yuan et al., 2021c)							*	
	PaCa-ViT (Grainger et al., 2023)		X			X		*	
	FLatten (Han et al., 2023)		X		*				
	Shunted-SA (Ren et al., 2022)					X		*	
	VSA (Zhang et al., 2022d)				X	X			*
	NA (Hassani et al., 2023)					X			*
	T2T (Yuan et al., 2021c)							*	
	RegionViT (Chen et al., 2022a)					X		*	X
Small	UniFormer (Li et al., 2023c)		*			X		X	
	Wave-ViT (Yao et al., 2022)		X		*	X			
	BiFormer (Zhu et al., 2023)				X	X		*	X
	MiniViT (Zhang et al., 2022c)				*	X			
	DaViT (Ding et al., 2022)						*		X
	VAN (Guo et al., 2023)		*			X		X	
	XCiT (Ali et al., 2021)	X		*					
	DGA (Liu et al., 2022)							*	X
	LS-Attn (Zhu et al., 2021)						*	X	X
	DAT (Xia et al., 2022)				*				
Base	VOLO (Yuan et al., 2022)				*		X		X
	Refiner (Zhou et al., 2021b)		*		X				
	Slide (Pan et al., 2023)		X			X			*
	FasterViT (Hatamizadeh et al., 2024a)		X			*			X
	CSWin Dong et al. (2022)	*	X						X
	Vicinity (Sun et al., 2023)				*				X
	QuadTree (Tang et al., 2022a)				X			*	
	Contextual (Li et al., 2022d)	*						X	
	MViT (Fan et al., 2021)					*			
	MSG (Fang et al., 2022)							X	*
Large	LV-ViT (Jiang et al., 2021b)							*	
	MaxViT (Tu et al., 2022)	X	X			X	*		
	GCViT (Hatamizadeh et al., 2023)		*			X			X
	DGT (Liu et al., 2022)		X			X		*	X
	CaiT (Touvron et al., 2021b)			X				*	
	DeepViT (Zhou et al., 2021a)				*				
	Twins (Chu et al., 2021)							X	*
	Scalable-ViT (Yang et al., 2022b)				X	X	*		X
	RVT (Mao et al., 2022)				*			X	
	ConViT (d'Ascoli et al., 2021)	*		X			X		

coalition has provided a robust way to achieve locality and a better spatial feature extraction while retaining global context. Numerous studies have focused on manipulating ViT by replacing some modules with convolutional layers or adding these layers to a pre-existing architecture.

The first work that appeared to use convolutional layers with a ViT architecture is Convolution Vision Transformer (CvT) (Wu et al., 2021). It replaces ViT's non-overlapping patch embedding with overlapping convolutions across multiple stages, enabling spatial downsampling and richer feature hierarchies like CNNs. It further uses depth-wise separable convolutions instead of standard linear projection in self-attention, to capture local context while reducing computational cost. It also eliminates positional encoding as convolutions inherently model spatial relationships, simplifying design for variable resolution inputs.

ResT (Zhang and Yang, 2021) presents a memory-efficient multi-scale vision Transformer that compresses memory using depthwise convolution. The authors present that each head in the vanilla ViT is only responsible for a fixed-sized subset of input feature maps which degrades performance. To solve this issue, they introduced Efficient Multi-Head Self-Attention (EMSA) where the spatial dimension is re-shaped from 2D to 3D and fed to a depthwise convolution to reduce the height and width dimensions by an adaptive factor. This 3D token is returned to 2D and fed to two sets of projections to get key and value vectors to compute the attention score. To capture more low-level information, different patch embedding and positional encoding

techniques are proposed, which are shown in Fig. 4(fourth from left). For patch embedding, three convolution layers are used where the spatial dimension is downsampled by a factor of 4 and the channel dimension is increased by 2. In the positional encoding, a spatial attention module named pixel-attention (PA) is presented to encode positions. Specifically, PA applies a depthwise convolution operation to get the pixel-wise weight and then rescales with a sigmoid function. Since the input token in each stage is also obtained by a convolution operation, the positional encoding is embedded into the patch embedding module to improve the performance. This technique demonstrates the potential of being a backbone for dense prediction.

Convolutional Neural Networks Meet Vision Transformers (CMT) (Guo et al., 2022a) integrates convolutional layers in different parts of a ViT, starting from a convolutional stem for initial feature extraction. A local perception unit extracts local context within patches through depthwise convolution in the CMT block. Additionally, the CMT block includes a Lightweight Multi-head Self-attention module that applies depthwise convolution to key and value matrices before the attention operation, reducing the computational cost.

Lite ViT (LVT) (Yang et al., 2022a), designed for mobile, introduced Convolutional Self-Attention (CSA) and Recursive Atrous Self-Attention (RASA). CSA integrates local self-attention into a convolution kernel in the first stage to enrich low-level features. In the last three stages, RASA incorporates multi-scale context when calculating the similarity map through depthwise convolutions and uses a recursive mechanism to increase representation capability with minimal extra parameters.

ConViT (d'Ascoli et al., 2021) combined a convolutional inductive bias with self-attention and developed a gated positional self-attention where Adaptive Attention Span (AAS) and Positional Gating are introduced. AAS trains only embeddings that control attention centers and spans while leaving relative positional encodings. This helps reduce the number of parameters while retaining long-range information. Positional Gating uses a learnable parameter to balance positional and content-based attention, avoiding dominance of either term. It allows the model to learn the importance of both forms of attention, improving performance and flexibility.

Li et al. (2023e) proposed EfficientformerV2 that injects local information into the value matrix of the multi-head self-attention module and enables communication between different attention heads through fully connected layers. Before applying attention, they added a Stride attention which downsamples the query, key and value matrices to a fixed spatial dimension and interpolates the output to the original resolution at earlier stages. Dual-path attention downsampling is also introduced in the downsampling stages using depthwise convolution, pooling and attention downsampling to wield both local and global dependency. These techniques help to reduce the latency and model size while preserving accuracy. This model is based on their previous EfficientFormer (Li et al., 2022e) model which is also based on convolutional layers.

These approaches integrate convolution into token embedding, attention projection, or downsampling stages, preserving global receptive fields while injecting spatial inductive bias. By incorporating depthwise convolutions and convolutional stems, they stabilize training, improve sample efficiency, and often eliminate or replace the explicit positional encoding. Additionally, through spatial reduction or lightweight attention, they mitigate the quadratic complexity of self-attention, making them better suited for dense prediction and resource-constrained settings than the original ViT.

Glance-and-Gaze ViT (Yu et al., 2021) proposed two branches in the architecture, the Glance branch and the Gaze branch. The Glance branch works as a vanilla self-attention, similarly to ViT, where the Gaze branch performs depthwise convolution. It was concluded that direct down-sampling of the input causes information loss. Therefore, the input was changed for Glance by splitting feature maps into several partitions. These partitions are then converted to query Q, key K and value V vectors. Q and K vectors are used for self-attention, while vector V, is sent to the Gaze branch for convolution operations. The Gaze branch compensates for the local patterns by applying depthwise convolution, which is missed in Glance. At the end, the results from both branches are merged and sent to MLP layers.

EdgeViT (Pan et al., 2022a) is designed to work on mobile devices. It involves a Local–Global–Local (LGL) bottleneck for information exchange between different layers by integrating self-attention and convolutions. The design uses depthwise and pointwise convolutions for local aggregation where local information from neighboring tokens is first aggregated to the target token. Then, self-attention is applied to a sparse set of delegate tokens, instead of all tokens, to gather long-range information. Updated global information from delegate tokens is propagated via transposed convolutions to non-delegate tokens in the local neighborhood. It reduces the number of tokens involved in attention, thereby improving efficiency.

Next-ViT (Li et al., 2022b) involves the Next Convolution Block (NCB) to capture local information using convolutions in a novel Multi-Head Convolution Attention and the Next Transformer Block (NTB) to collect global context using self-attention. These blocks are combined using the Next Hybrid strategy (NHS), which stacks NCB and NTB using a *(N+1)*L* hybrid paradigm. This means that for each stage, there are N instances of the NCB followed by 1 instance of the NTB and repeated L times across different stages. It allows the model to mix high and low-frequency signals for better representation learning.

UniFormer (Li et al., 2023c) uses a local Multi-Head Relation Aggregator (MHRA) to capture local token affinity in the shallow layers using a small learnable parameter matrix, where the parameter matrix acts similarly to a convolution filter. In the deep layers, it employs a global MHRA that learns global token affinity through similarity comparison, similarly to vanilla self-attention, to capture long-range dependencies. This maintains the benefits of self-attention while addressing its computational drawbacks.

The Global Context Vision Transformer (GCViT) (Hatamizadeh et al., 2023) alternates local and global self-attention modules to extract spatial features. At each stage, local self-attention operates within fixed windows while global self-attention accesses global features generated by a CNN-like query generator which processes the entire image once per stage. This allows interaction between local and global regions, addressing the limitation of local attention. The global attention is computed using global query tokens which are generated using a modified Fused-MBConv(Mobile Inverted Bottleneck Convolution (Sandler et al., 2018)) block inspired by EfficientnetV2 (Tan and Le, 2021) and combine them with local keys and values, improving accuracy.

Inception Transformer (Si et al., 2022), inspired by Inception (Szegedy et al., 2015), introduced a mixer that splits the input features along the channel dimension and feeds the split components into high-frequency and low-frequency mixers. The high-frequency mixer consists of convolution and max-pooling operations while the low-frequency mixer utilizes self-attention with an average pooling layer to reduce spatial scale. It allows the model to capture detailed local information and global information effectively.

In contrast to projection-level fusion, these architectures generally decouple local and global modeling into separate pathways or alternating stages. Convolution and self-attention are often treated as complementary operators, either sequentially stacked, split into parallel branches, or integrated within hybrid bottlenecks to facilitate controlled information exchange between fine-grained local aggregation and long-range dependency modeling. In some designs, computational efficiency may be improved by restricting global attention to selected tokens, reduced spatial resolutions, or specific stages of the network. However, such approaches often introduce additional architectural complexity, and overall performance may depend on how convolutional and attention blocks are allocated and scheduled throughout the model.

Refiner (Zhou et al., 2021b) introduced attention expansion and distributed local attention. Attention expansion projects the attention

maps to a higher-dimensional space to increase their diversity. Distributed local attention applies convolutions to attention maps to capture local details. This approach addresses the over-smoothing issue of vanilla self-attention.

CoaT (Xu et al., 2021a) was introduced to enhance multi-scale and contextual modeling capabilities. It involves two mechanisms, Co-scale and Conv-Attentional mechanisms. Co-scale mechanism maintains the integrity of the Transformer encoder branches at different scales while allowing cross-scale communication. It is achieved through a series of serial and parallel blocks, enabling fine-to-coarse, coarse-to-fine, and cross-scale interactions. The Conv-Attentional mechanism realizes relative position embeddings using an efficient convolution-like implementation in the factorized attention module. It approximates the softmax attention map using two linear functions $\phi(\cdot)$ and $\psi(\cdot)$ and computing keys and values together which helps the model to handle high-resolution images. It combines convolutional relative position encoding and convolutional position encoding to improve computational efficiency and modeling capability.

Multi-scale High Resolution ViT (HRViT) (Gu et al., 2022) uses a multi-branch high-resolution architecture that extracts features at different scales in parallel with cross-resolution fusion, enhancing the model's ability to capture multi-scale elements. It uses augmented self-attention to remove redundant keys and values and boosts expressiveness by merging parallel convolution paths to inject inductive bias, extra non-linearity units and shortcuts to increase the feature diversity. It also adopts a multi-scale convolutional feedforward network to capture local context at different scales. Finally, its high-resolution convolutional stem and patch embedding help retain fine-grained details and facilitate cross-scale interaction while reducing computational cost.

Contextual Transformer (Li et al., 2022d) integrates context mining and self-attention into a unified structure, unlike self-attention which handles pairwise query–key relationships independently. This Transformer applies group convolution to contextualize keys, leading to a static representation of inputs. It further concatenates keys with queries to learn a dynamic multi-head attention matrix. Finally, both static and dynamic contexts are adaptively fused to produce the final output.

Wang et al. (2023b) proposed the Evolving Attention Mechanism which also establishes communication between attention maps by planting a skip connection. This facilitates the information flow of attention patterns. They start with capturing the evolution of attention maps across layers, leveraging convolutional layers to refine these maps. The output attention map of one head is combined with a convolution and passed to the next head. They use convolutional inductive biases to improve the quality and explainability of attention maps.

The integration strategy of these approaches focuses on refining the attention mechanism itself using convolutional operators. Instead of injecting locality before or after attention, some of these methods alter how similarity maps are constructed, propagated, or regularized. Convolution acts as a structural prior that reshapes attention distributions or contextualizes the key–query exchange, potentially mitigating over-smoothing and improving feature diversity. This approach suggests that convolution can function not only as a feature extractor but also as a regulator of attention dynamics. However, while these architectural innovations often improve performance and map distinctness, they increase structural complexity and may require more sophisticated visualization techniques to interpret the modified attention weights.

CieT (Yuan et al., 2021a) integrates CNN into its three different modules to enhance the local feature extraction while preserving the global receptive field. The image-to-tokens module extracts patches from low-level features instead of raw images, improving feature representations. The locally enhanced feed-forward module replaces the standard feed-forward network with a depthwise convolution to promote spatial locality by capturing local correlations among neighboring tokens. The layer-wise class token attention utilizes multi-level representations to refine the final output through multi-level feature maps. These modifications combine CNN's local pattern recognition strength with the Transformer's global context recognition to improve performance.

Restormer (Zamir et al., 2022) applies self-attention to channel dimension instead of spatial dimension. The model computes cross-covariance across channels to generate an attention map to obtain global information. They also use depthwise convolution to emphasize the local context. This approach enables the model to handle the restoration of high-resolution images efficiently.

Guo et al. (2023) proposed a Large Kernel Attention (LKA) that consists of three components, a spatial local convolution, a spatial global convolution and a channel convolution using depthwise convolution, depthwise dilation convolution and pointwise convolution (1×1 convolution), respectively. This composition allows the model to capture long-range context while maintaining a local structure with minimal computational cost.

While these models incorporate convolution at the module level to enhance spatial inductive bias while maintaining or reformulating global context modeling. Rather than separating branches or directly modifying attention maps, they integrate convolution into core components such as tokenization, feed-forward layers, or as a complete replacement for standard attention operators. This can improve local feature aggregation and efficiency, though it introduces varying trade-offs in complexity and interpretability.

Advantages and limitations

The fusion of convolution and self-attention bridges the global power of self-attention and the local power of convolution. This helps content-adaptive context modeling through the incorporation of local inductive bias which leads to improved sample efficiency and faster convergence. Convolution helps to handle fine-grained details, and the fusion with attention can mitigate the aforementioned complexity issues, especially on dense tasks.

However, they offered limited architectural flexibility and only marginal gains once attention mechanisms matured. Early fusion helped introduce locality bias and stabilize training, but it often increased computational redundancy through overlapping local modeling in convolution and attention, without resolving core scalability challenges such as quadratic complexity. Further, early models retained a flat token structure and did not fundamentally address the quadratic complexity of global attention.

In contrast, hybrid models, described in the next sections, such as Pan et al. (2023), integrate convolution as independent stages or parallel branches alongside global attention, allowing each component to operate in its optimal regime. This design improves scalability to high resolutions, reduces computation, and better supports dense prediction through explicit multi-scale modeling. At the same time, it preserves the global reasoning strength of Transformers while introducing structural inductive biases in a modular and efficient manner. As a result, the research community moved toward hybrid architectures that offer clearer efficiency gains, improved scalability, and stronger performance across diverse vision benchmarks.

Cui et al. (2022) and Huang et al. (2021), often employ both local mixing with convolutions and window attention, which adds more computation without clear gains. Moreover, Zhu et al. (2024b) suggests that MHSA at the window level is unnecessary next to convolution stacks. Even with convolution improving local context, when combined with self-attention, there is still a quadratic time and memory constraint, with the added complexity of the convolution, limiting real-time use on edge devices. Pan et al. (2022b) shows that fusion may also increase hyperparameter sensitivity, adding decisions on where to switch from convolution to attention or how to balance learnable mixing weights, requiring costly re-tuning.

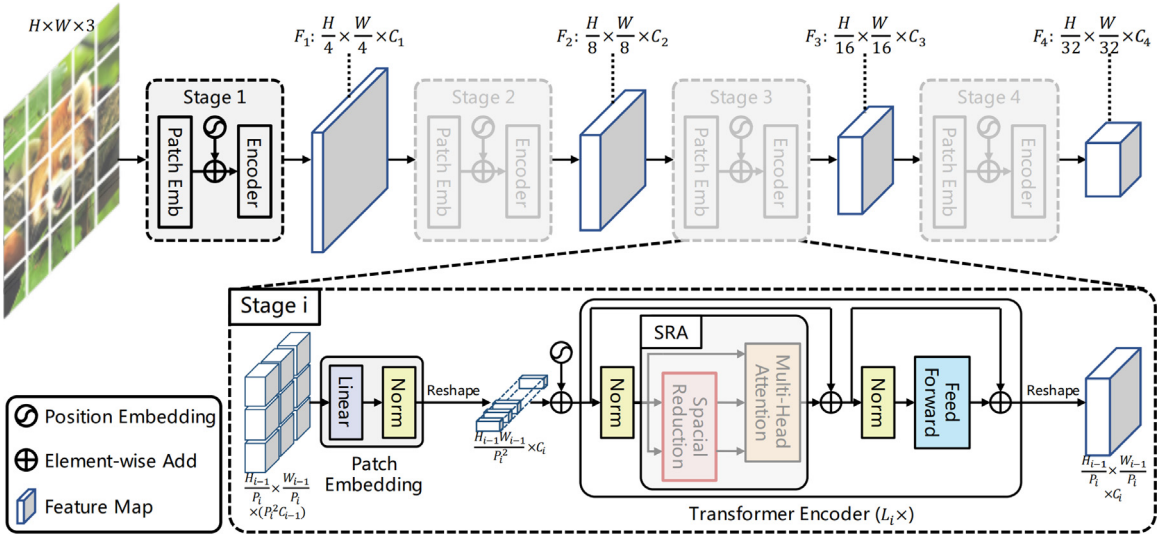


Fig. 3. Overview of the Pyramid Vision Transformer (PVT) architecture taken from the original work (Wang et al., 2021). Top: The model processes images through multiple stages with decreasing spatial resolutions. Bottom: Transformer encoder with Spatial Reduction Attention (SRA) which builds a hierarchical feature representation.

3.2. Hierarchical attention

Hierarchical attention emerged as a structural response to a fundamental limitation of vanilla ViT of treating all tokens at a single resolution uniformly. In contrast to flat architectures, hierarchical models progressively reduce spatial resolution while increasing channel capacity, thereby introducing a multiscale inductive bias similar to convolutional pyramids. This design simultaneously addresses mitigating quadratic attention cost at high resolutions and enabling feature abstraction across scales for dense prediction tasks. Unlike convolution-fused models 3.1, which add locality to flat architectures, hierarchical attention restructures the Transformer into a pyramid of progressively coarser representations, fundamentally reorganizing global token interactions.

The Pyramid Vision Transformer (PVT) (Wang et al., 2021) was the first attempt to use a pyramid structure in ViT. It primarily focuses on pixel-level dense predictions using a multi-scale approach, with one goal being to process high-resolution images with less computational cost. It generates feature maps with different scales for dense prediction tasks. A progressive shrinking strategy is used to control the scale of feature maps using spatial-reduction attention (SRA). The architecture is split into four stages and every stage takes a different size of input. First, the feature maps are divided into 4 × 4 patch size, (*HxW/4x4*) patches, then the patches are flattened and projected to the channel dimension C ($C_1 = 64$). For the next stages, the input from the previous stage is used to generate outputs with 8, 16, and 32 patch sizes. The detailed architecture can be seen in Fig. 3. This reduction is implemented in the Transformer encoder block, where SRA is used to reshape the patch size instead of multi-head attention, reducing the computational cost. Being the first model in this category, it inspired many researchers to use it as a backbone or implement other novel hierarchical techniques on it.

Fan et al. (2021) introduced Multiscale ViT (MViT), creating a multiscale pyramid architecture by hierarchically expanding the channel dimension while reducing the spatial resolution across different stages. Pooling attention was presented in the architecture, where a pooling kernel is computed along each dimension across all the vectors, allowing flexible resolution modeling, reducing the spatial resolution. The final results show a fundamental advantage over single-scale VTs for image recognition. Later, MViT was further improved with a newer version (MViTv2) (Li et al., 2022c) where the absolute positional embedding is replaced with a decomposed relative position embedding to improve patch interactions. The residual pooling connections are also added to increase the information flow between the layers and facilitate the training of pooling attention. To improve the model for detection and segmentation tasks, Hybrid Window Attention (HWIN) and the Feature Pyramid Network (FPN) (Lin et al., 2017) are integrated into the model. HWIN combines pooling and window attention to add cross-window connections. It computes local attention within a window in all but the final three stages. HWIN's output is fed into the FPN which helps extract global information, achieving state-of-the-art performance in multiple tasks. The architecture can be seen in 4(third from left).

The Pyramid Pooling Transformer (P2T) (Wu et al., 2022b) adapted the idea of pyramid pooling from Lazebnik et al. (2006), He et al. (2015) and forwarded it to the Transformer for extracting rich contextual information. It incorporates the pyramid pooling in the MHSA block and uses multiple pooling operations with different receptive fields to generate a set of pyramid feature maps, reducing the length of image tokens. These maps are then used to compute attention which allows the model to capture multi-scale information, making the model efficient and powerful at capturing more details.

These Pyramid-based models progressively reduce spatial resolution while increasing channel capacity, forming hierarchical feature representations. Rather than operating on a flat token space, they reorganize computation across multiple scales, reintroducing a pyramid structure similar to that of CNN backbones. This design makes them well-suited for dense prediction tasks such as detection and segmentation. Resolution reduction improves computational efficiency and helps manage the quadratic complexity of attention, while the hierarchical structure provides multi-scale features. As a result, hierarchical Transformer architectures have become widely adopted backbones for dense vision tasks, demonstrating strong performance on large-scale benchmarks.

FasterViT (Hatamizadeh et al., 2024a) introduced hierarchical attention by incorporating carrier tokens that summarize local windows and propagate global information. First, carrier tokens are computed using self-attention to gather global context and then are concatenated with local tokens, allowing local windows to interact with carrier tokens to exchange global and local information by applying self-attention. Finally, these tokens are split and used in the subsequent hierarchical attention layers. This hierarchical attention technique splits full attention into local and sub-global attention and the carrier tokens help to achieve global context and facilitate information exchange.

The Sliced Recursive Transformer (Shen et al., 2022) improved the parameter utilization using a recursive operation where weights

are shared across the depth of the Transformer for better feature extraction and performance. To mitigate the additional computation associated with recursive operations, a sliced group attention technique is introduced. This technique decomposes the global self-attention into multiple slices, applying attention within each slice and recursively aggregating information. This reduces computational cost without sacrificing performance. This structure allows for extremely deep Transformers with over 100 or even 1000 shared layers while maintaining a compact size, achieving substantial gains in accuracy with fewer parameters.

Unlike pyramid-based spatial shrinking, these methods build hierarchy through structured token communication rather than heavy downsampling. Mechanisms like carrier tokens or sliced attention summarize and redistribute global context while largely preserving the spatial grid. This shifts hierarchy from resolution reduction to organized information flow, where global representations are progressively distilled and propagated. Such designs suit efficient or very deep models, showing that Transformer hierarchy can arise from coordinated token interactions as well as spatial scaling.

Nested Transformer designs like TNT (Han et al., 2021) introduce hierarchy at the granularity level rather than through spatial downsampling. Images are split into patches, then further divided into sub-patches. An inner Transformer models fine-grained interactions within each patch, while an outer Transformer captures global relationships across patches. This dual-level attention mirrors CNN-style feature hierarchies while retaining self-attention flexibility, improving accuracy by separating local and global reasoning. However, the added architectural complexity increases computational cost, limiting scalability compared to simpler pyramid-based designs.

Advantages and limitations

Hierarchical attention constitutes a structural evolution of VTs rather than a local modification of self-attention. This reorganizes representations across scales through spatial resolution reduction, token-level summarization of global context, and multi-granularity modeling with nested stages. Unlike ViT-based designs that preserve flat token interactions, hierarchical architectures introduce a multiscale inductive bias suited to dense prediction, which explains their dominance over earlier convolution–attention hybrids in backbone design.

By shrinking spatial resolution or restricting token interactions, hierarchical attention mitigates quadratic complexity, but this efficiency comes at a cost. Fine-grained information is discarded and must be reconstructed, typically through the use of skip connections or integrated Feature Pyramid Networks that bridge the gap between high-resolution early features and deep semantic representations. While these designs reduce computational complexity, memory usage can be uneven because keys and values are cached at each pyramid level. Moreover, theoretical results indicate that hierarchical stacking does not resolve the fundamental expressiveness limits of self-attention, as modeling certain periodic or deeply recursive structures still requires scaling heads or depth with input size. These architectures also inherit the interpretability limitations of standard Transformers. Overall, hierarchical attention offers an effective efficiency–accuracy trade-off, yet leaves open challenges in memory efficiency, interpretability, and expressive capacity.

3.3. Cross attention

Cross-attention extends self-attention by enabling interaction between distinct token sets or feature spaces. In VTs, it supports multi-scale fusion, semantic–spatial interaction, latent bottlenecks, and multi-branch communication. Unlike hierarchical attention 3.2, which reorganizes representations across scales, cross-attention maintains parallel token streams and explicitly models their exchange, making it well-suited for multi-resolution learning and dense prediction tasks that require feature alignment across abstraction levels.

The first work on multi-scale ViT was done by Chen et al. (2021), where the authors tried to improve the recognition accuracy for image classification by learning multi-scale features. They combined image patches of different scales using a fusion method based on cross-attention and proposed dual branch Transformers to combine the image patches to produce stronger image features. This method utilizes large and small patch tokens for bidirectional information exchange between two branches in linear time. This involves the classification token from one branch and patch tokens from the other branch, enabling cross-branch interaction. The classification token at each branch works as a mediator, transferring information from the opposite branch's patch tokens and reintegrating it into its own. Since the classification token already encodes abstract information from its branch, this information transfer facilitates multi-scale feature learning. Following fusion, the classification token re-engages with its own patch tokens in the next Transformer encoder, distributing the learned information to enhance their representations. The results indicate that this method achieves superior performance compared to several VTs and efficient CNN models.

Dual Vision Transformer (Yao et al., 2023) extracts global semantic information that serves as prior information to assist finer local feature extraction in a twin-pathway design — namely, the semantic pathway and the pixel pathway. The semantic pathway transforms the input feature map into semantic queries via self-attention and extracts the semantic tokens. These tokens are then used in pixel pathways in the form of keys/values and perform multi-head attention to refine input feature maps via cross-attention. In the end, self-attention is performed again over concatenated semantic and local tokens. Henceforth, reducing the computational complexity without compromising much accuracy.

Wang et al. (2022c) proposed CrossFormer cross-scale attention where they added a cross-scale embedding layer to integrate information from different scales into tokens to enhance cross-scale descriptiveness. Additionally, they introduced long-short distance attention which divides self-attention for small and large-scale features to reduce complexity. Later, they enhanced their technique (Wang et al., 2023a) to handle large-sized attention maps and the amplitude explosion of each block. They assign smaller group sizes in shallow layers for local features and larger groups in deep layers for global attention to reduce computational costs in earlier layers and ensure effective global attention in later layers. To control the amplitude, they used depth-wise convolution and normalization without any residual connection, bringing a performance gain in their model.

These models show that cross-attention serves as a sophisticated mechanism for multi-scale feature integration. Rather than forcing features into a single resolution pyramid early in the network, these architectures maintain parallel streams and utilize attention as a structured channel for information exchange. This approach reduces premature information compression while enabling adaptive fusion. Compared to convolution-based fusion, cross-attention offers content-adaptive integration rather than fixed kernel aggregation. While these multi-stream interactions introduce architectural complexity, they often incorporate specialized attention patterns such as mediator tokens to maintain computational efficiency while capturing both fine-grained details and broad semantic context.

CAT (Lin et al., 2022) presents a hybrid interpretation of cross-attention at the patch level by decoupling intra-patch and cross-patch interactions. Specifically, it introduces Inner-Patch Self-Attention (IPSA) to model local information within each patch and Cross-Patch Self-Attention (CPSA) to capture global context by partitioning channel feature maps into patches and applying self-attention across them with limited computational overhead. This channel-wise partitioning approximates hierarchical, multiscale reasoning without explicit spatial downsampling or pyramid restructuring, enabling scale-adaptive feature generation for diverse vision tasks. However, by simulating multiscale modeling through structured channel partitioning, CAT blurs

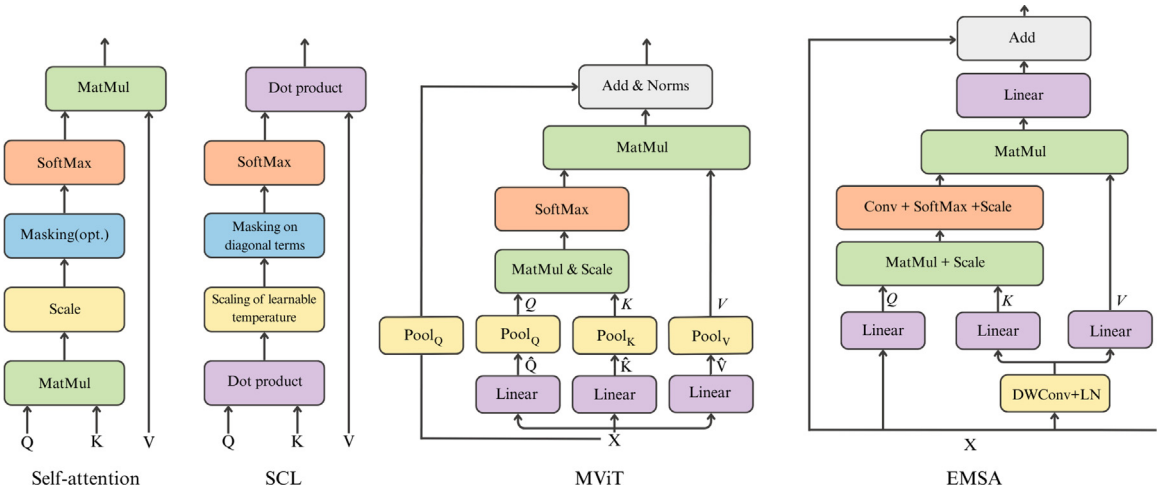


Fig. 4. Architectures of self-attention and its variants, from left to right: Standard Self Attention, ViT for small-size dataset (SCL), Multi-Scale Vision Transformer (MViT) and Efficient Multi-Head Self-Attention (EMSA). They illustrates how different methods introduce modifications or additional layers into the architecture, such as altering the scaling mechanism, adding pooling layers, or incorporating convolutional operations, to enhance efficiency, scalability, and representational capacity.

the distinction between cross-attention and structured self-attention, complicating direct comparative evaluation.

Perceiver (Jaegle et al., 2021) is designed to handle high-dimensional data using a latent array and iterative cross-attention. It processes the input using cross-attention to project a high-dimensional input array to a lower-dimensional latent space. This latent array acts as an attention bottleneck, allowing the model to iteratively refine the input by switching between cross-attention and self-attention blocks. This helps the model to focus on the most relevant parts of the input without having any size constraints. The Perceiver directly applies attention to the raw inputs and uses a latent space to handle the quadratic scaling problem of standard Transformers.

BiXT (Hiller et al., 2024) allows simultaneous attention between latent variables and input tokens, promoting an efficient bidirectional exchange of information which is dissimilar to conventional self-attention that is unidirectional. This method makes the attention more computationally feasible for longer input sequences.

ViT-adapter (Chen et al., 2023b) tried to improve ViT for dense prediction tasks. It uses an adapter module to inject inductive bias in ViT without modifying the original architecture. There are three modules, a spatial prior module to capture local semantics from the input image, a spatial feature injector to incorporate these spatial priors into the ViT, and a multi-scale feature extractor to reconstruct the multi-scale features. Cross attention is employed in these modules which allows the model to selectively focus on different parts of the input and ViT features, effectively combining the strengths of both convolution and Transformers.

Across these models, cross-attention evolves from a simple fusion operator into a flexible interface. In Perceiver and BiXT, cross-attention acts as a computational bottleneck, projecting high-dimensional inputs into a compact latent space to decouple model depth from input resolution. In adapter-based designs, it serves as a modular bridge, injecting localized inductive biases into a frozen backbone for task-specific tuning without altering the Transformer architecture. As either a scalability filter or spatial injector, cross-attention reshapes information flow across data scales while keeping internal processing cost fixed, though the trade-off between latent capacity and feature resolution remains design-dependent.

Advantages and limitations

Cross-attention enables structured interaction between distinct token sets. This serves multi-scale feature alignment, latent bottleneck projection for improved scalability, and inductive bias injection for task adaptation. Unlike hierarchical attention in Section 3.2, which enforces progressive compression, cross-attention preserves parallel representations and supports dynamic, content-aware fusion. This flexibility makes it well-suited for dense prediction, multimodal learning, transfer learning, and unified multi-scale processing, where it often outperforms simple concatenation while preserving local detail.

Despite these advantages, cross-attention is rarely the dominant backbone strategy in vision models. Maintaining multiple token streams increases memory usage and complicates kernel optimization, and the quadratic cost of attention becomes prohibitive for large or multi-scale inputs. Although latent bottleneck approaches mitigate scaling issues, they introduce compression hyperparameters that are difficult to standardize across tasks. Cross-attention also increases architectural complexity and alignment sensitivity, since mismatched resolutions, semantic inconsistencies, or data discontinuities can degrade attention quality and overall performance. Furthermore, the entanglement of information across streams complicates interpretability and model development. Improving efficiency and robustness when handling heterogeneous, multimodal, and multi-scale data, therefore, remains an important research direction.

3.4. Window-based self-attention

The original self-attention retrieves global information decently but suffers from the high computational cost, making it ineffective in handling large-scale patches. Window-based attention emerged as a powerful strategy to limit computational resources by restricting attention to localized windows or regions and retaining accuracy, making it one of the most widely adopted strategies for scaling VTs to practical solutions.

Swin Transformer (Liu et al., 2021) was proposed to solve fixed scaling and inefficiency to handle high-resolution data, along with the quadratic complexity of the vanilla ViT. In this method, the architecture is split into multiple stages. A patch merging technique is introduced that reduces the spatial dimension while increasing the channel dimension of feature maps. Features of each patch with 2×2 neighboring patches are concatenated and the output is sent to the Swin Transformer block. In this block, shifted window-based self-attention is used which can be seen in Fig. 5. Initially, self-attention is applied to the standard 2×2 window partitions, after which the windows are shifted by 2×2 pixels. This shift introduces connections between

different windows, which are absent in the standard window partitions. However, this process creates more windows than the original 2×2 partitions. To address this, they employed a cyclic shift technique and used masked multi-head self-attention (MMHSA) to ensure that attention computations remain confined within each window. They achieve state-of-the-art performance while reducing the attention complexity from quadratic to linear.

The Progressive Shift Ladder Transformer (PSLT) (Wu et al., 2023) extended the Swin Transformer through a new ladder self-attention with multiple branches. The ladder self-attention block divides the input feature map into multiple branches along the channel dimension to improve local attention. It adopts a windows-based progressive shift mechanism (PSW) that enlarges the receptive field by computing similarities between pixels in each branch, unlike Swin which stacks an even number of window-based blocks. Initially, features are processed without shifting in one branch, while the other branches apply these shifts to enable local interactions. The output of each branch is progressively transmitted to the next, with the PSW-MHSA using both the previous branch's output and the current input for self-attention. It expands the receptive field while improving computational efficiency and achieving comparable performance.

Shuffle Transformer (Huang et al., 2021) also extended the Swin Transformer with a new technique, the spatial shuffle. In Swin, the similarity is only calculated within windows where the spatial shuffle obtains data from different windows. This reshapes the attention map into a new dimension and transposes it as input for the next layer by putting the tokens from distant windows together and building connections between windows. These tokens are then adjusted back to the original positions by spatial alignment (the inverse of the spatial shuffle). To solve the grid issue a convolution layer with a residual connection is considered. The proposed allowed information to flow between all windows, improving performance with similar computational complexity.

Twins (Chu et al., 2021) uses a pyramid vision Transformer (PVT) (Wang et al., 2021) as a base and uses a positional encoding generator to replace the absolute encoding in PVT. In local self-attention, they divide feature maps into sub-windows and perform operations locally. For global communication between sub-windows, extra global self-attention layers are used which can help to exchange information between windows. From each sub-window, a single representative was selected to communicate with other sub-windows which serves as the key in self-attention. This reduced cost and improved performance through token-interaction optimization.

High Resolution Transformer (HRFormer) (Yuan et al., 2021d) took advantage of the multi-resolution parallel design introduced in HRNet (Wang et al., 2020), along with local-window self-attention that performs self-attention over small non-overlapping image windows, improving computation efficiency.

To enhance ViT's capability to perform for high-resolution images, Messenger (MSG)-Transformer (Fang et al., 2022) was proposed. The addition of an MSG token was used to share information between different local windows. Input features are divided into multiple windows and then each window is aggregated with an MSG token. In the MHSA block, attention is performed on patches including the MSG token as shown in Fig. 5. These MSG tokens from different windows are then shuffled and reshaped, exchanging information between different windows.

These architectures address the scalability limits of vanilla ViT by combining local window attention with cross-window communication. Through hierarchical designs such as Twins and PSLT and parallel high-resolution streams like HRFormer, they reduce quadratic global attention to sub-quadratic or linear complexity. By preserving high-resolution features and integrating well with pretrained frameworks, window-based Transformers have become preferred backbones for dense prediction tasks like detection and segmentation, often surpassing hybrid convolution–attention models in both accuracy and efficiency.

Li et al. (2021) is another technique inspired by the Swin Transformer that enables local to global information exchange at each stage. Similar to Swin, the attention maps are calculated within each window. The window size is set equal to a decided smallest feature scale which enables the model to reason globally. After each Shifted-window MSA module, the downsampled maps are upsampled to the same resolution and combined before being fed into further layers. This gathering of local and global context within the same blocks improved accuracy with extra computational overhead.

Researchers have been trying to reduce the cost and improve model performance by applying either coarse-grained global attention or fine-grained local attention. Both options lead to sub-optimal solutions. Focal Attention (Yang et al., 2021) was proposed to incorporate both fine-grained local and coarse-grained global tokens with the goal of making Transformers suitable for high-resolution input. Instead of attending to all tokens at a fine-grained level, it attends to the fine-grained tokens locally and the summarized tokens are processed globally. However, extracting surrounding tokens for all queries is computationally expensive. To overcome this, window-based self-attention is used which extracts the surrounding tokens for each query token in the feature map inside the window. As tokens inside a window share the same set of surroundings, the self attention calculation is simpler, improving performance.

MixFormer (Chen et al., 2022b) combines local-window self-attention with depthwise convolution in a parallel design to model cross-window connections and expand receptive fields. Bi-directional interactions are used across branches to provide additional information in both channel and spatial dimensions, leading to improved performance.

Zhang et al. (2023) also incorporate depthwise convolution with window-based multi-head self-attention and balance the parameters to make the model suitable for mobile devices.

Uformer (Wang et al., 2022d) inspired by U-Net (Ronneberger et al., 2015) constructs a Locally-enhanced Window Transformer block, performing non-overlapping window-based self-attention to capture local context instead of global attention for restoring images. It uses a depthwise convolutional layer to extract local information which helps improve accuracy.

Slide Transformer (Pan et al., 2023) reinterprets the inefficient *Im2Col* function to a new row-based perspective and replaces it with depthwise convolutions. They combine the local inductive bias of convolutions with the dynamic feature selection of self-attention. They also include a deformed shifting mechanism to relax fixed positions of keys and values within the local region using a designed kernel weight, as shown in Fig. 5. This performs shifting toward different directions with the learnable parameters to enhance flexibility and achieves an improved trade-off between computational efficiency and performance.

These methods highlight that window-based attention benefits from mechanisms capturing fine-grained local correlations, leading to the reintroduction of convolution. Unlike models in Section 3.1, convolution here acts as local refinement within a windowed framework rather than a dominant global bias. Thus, windowing governs global interactions, while convolution enhances local representations without replacing the Transformer backbone.

HiLo Attention (Pan et al., 2022c) divided the self-attention mechanism into two parts, high and low frequency attention. Hi-Fi attention captures local details and Lo-Fi obtains global information within windows, while the low-frequency data is obtained using average pooling. This splits the heads of MHSA into two groups for high and low attention which are concatenated at the final layer, achieving improved efficiency, especially on high-resolution images.

Zhang et al. (2022d) introduced Varied-Size Window Attention (VSA) that predicted the target window size and location using a reference window as in Fig. 5. They compute key and value tokens using sample features of each attention head's target windows, promoting information exchange. Later, they extended their model to another Quadrangle Attention (Zhang et al., 2024) that includes the

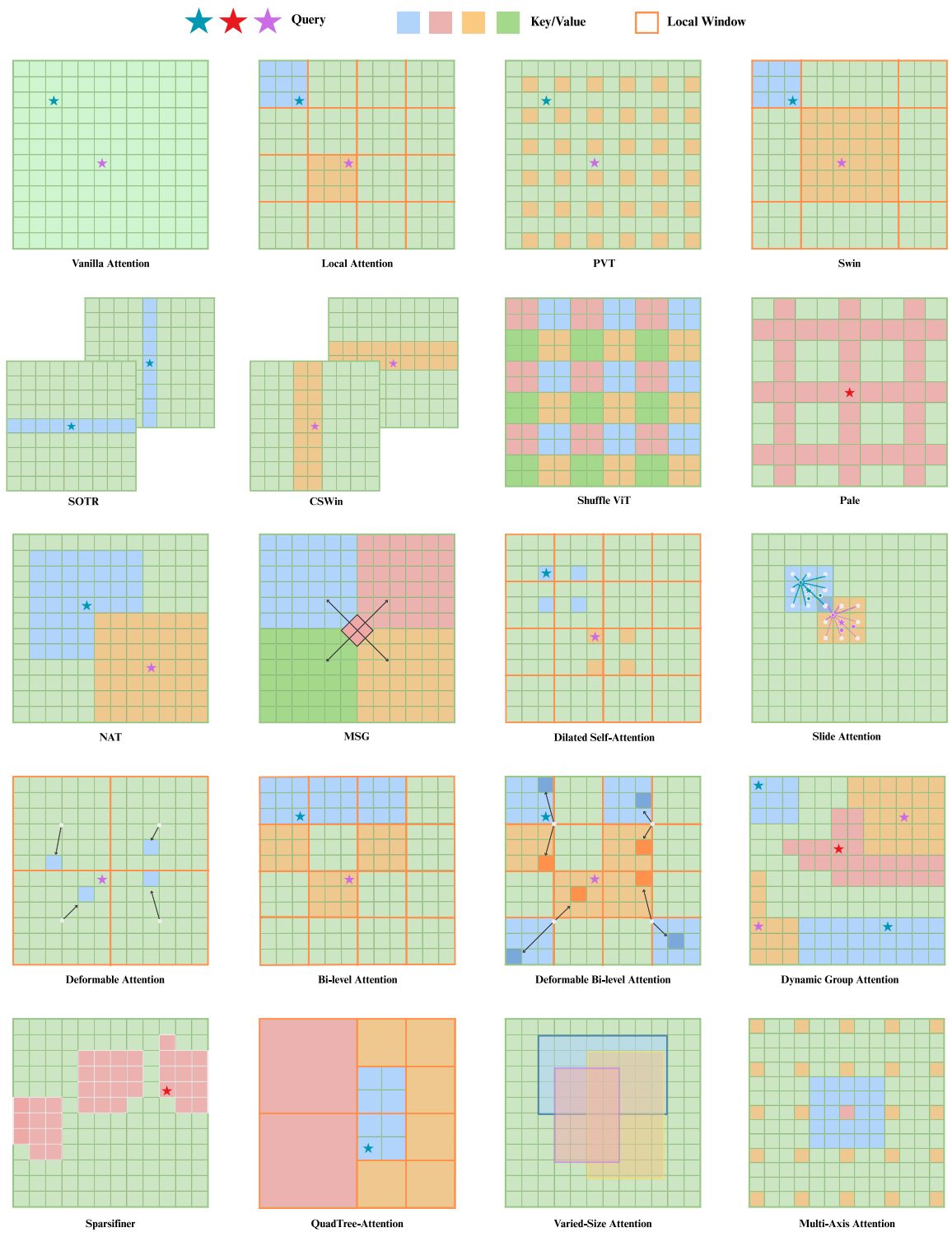


Fig. 5. Visualization of attention mechanisms in Vision Transformers, illustrating how queries (stars) attend to key/value tokens (colored regions) across spatial configurations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

prediction of the target window orientation and shape using an end-to-end quadrangle module. This allowed their method to remove the fixed-size window restriction found in the original architecture and adopt variable, data driven transformations.

Neighborhood Attention (NA) (Hassani et al., 2023) tried to calculate local attention in more detail using less computational resources. This considers Swin's shifted window technique, intended to expand the receptive field to capture more local information, as one of the fastest approaches but ultimately find it inefficient. In vanilla ViT, self-attention applies directly to patches while Swin first groups patches into windows and then shifts these windows to increase the receptive field. However, both techniques are considered inefficient. Hassani et al. (2023) claims that the most direct way to expand the receptive field is to allow each pixel to attend to its neighboring pixels which can restrict self-attention locally. Therefore, it performs a pixel-wise operation, localizing self-attention to the nearest pixels. This pipeline

also consists of a sliding windows mechanism which enables pixels to adapt to dynamically shifting windows, allowing the expansion of the receptive field as shown in 5. This is implemented using the dot product of Query Q with its nearest neighboring Key K along with a positional Bias. This NA is wrapped in a NAT block, which embeds inputs using convolutions, improving accuracy and efficiency.

DilateFormer (Jiao et al., 2023) balances computational efficiency and long-range dependency modeling by introducing Multi-Scale Dilated Attention (MSDA), shown in Fig. 5. MSDA sparsely samples keys and values within a dilated sliding window, capturing local and sparse interactions efficiently. They use different dilation rates per attention head to model multi-scale dependencies without extra parameters, achieving competitive performance with a significant reduction in FLOPs.

Adaptive or flexible local attention methods aim to address limitations of fixed-window partitioning by allowing receptive fields to vary across heads, dilation rates, or predicted window configurations. Instead of relying solely on static window shapes, some designs introduce learned window sizes, orientations, or multi-scale sampling strategies to expand receptive fields and mitigate boundary fragmentation. Techniques such as sparse multi-scale sampling and frequency-based decomposition offer a more refined balance between local detail and global context. While these approaches increase modeling flexibility, they may introduce additional implementation complexity compared to fixed-window schemes. The practical trade-offs between adaptability, efficiency, and hardware compatibility influence their adoption in large-scale vision backbones.

Advantages and limitations

Window-based attention offers a pragmatic compromise between scalability, locality, and architectural simplicity. By restricting self-attention to local windows, it reduces computational and memory complexity toward linear scaling while reintroducing critical inductive bias such as spatial locality and translation invariance. Unlike aggressively downsampling hierarchical pyramids as in 3.2, early spatial shrinking is avoided and finer control is maintained over locality–globality trade-offs. Hierarchical windowing further introduces multi-scale representations through progressive patch merging and window adaptation, inheriting CNN-like multi-scale priors. These properties enable stable large-scale training, seamless integration with feature pyramid networks, and efficient use of GPU-friendly attention kernels, which supports the widespread adoption of Swin and Swin-like models as general-purpose vision backbones.

However, the same locality that ensures efficiency also limits expressiveness. Fixed window partitioning can fragment semantic objects, introduce boundary artifacts, and prevent tokens in non-adjacent windows from interacting within a single layer. Long-range reasoning, therefore, becomes indirect and depth-dependent, requiring stacked layers or shifted/adaptive windows to enlarge the effective receptive field. Such mechanisms mitigate but do not eliminate the issue, and progressively reintroduce computational overhead and architectural complexity, revealing that windowing postpones rather than fundamentally resolves the trade-off between global context and local detail. Moreover, rigid partitioning can be sensitive to resolution changes between training and inference, further exposing the fragility of fixed locality assumptions.

Overall, window-based attention is an effective engineering solution that balances efficiency and performance, yet it does not fully solve the core challenge of achieving efficient global reasoning without sacrificing locality or scalability.

3.5. Axial self-attention

Axial self-attention represents a factorization strategy that decomposes two-dimensional global attention into sequential one-dimensional operations along spatial axes. Instead of restricting attention to local windows as in Section 3.4 or shrinking resolution hierarchically as in Section 3.2, axial attention preserves global receptive fields while reducing computational complexity through directional decomposition. By computing attention along rows and columns separately, it achieves near-linear scaling with respect to spatial dimensions while retaining structured long-range interactions. This approach can be interpreted as a middle ground between full global attention and strictly local window-based attention.

Segmenting Objects with Transformers (SOTR) (Guo et al., 2021) is an early example, employing twin attention that decomposes the self-attention into independent rows and columns, reducing complexity and aggregating the contextual information between elements horizontally and vertically. Two different models are introduced in this paper, one of which only replaces self-attention with twin attention, the other combines twin attention with convolutional layers into a Transformer block to extract low-level features. Both techniques allow the model to capture local and global information more efficiently and improve their performance.

Cross-Shaped window self-attention (CSWin) (Dong et al., 2022) applies an axis-based self-attention mechanism, where the input feature map is split into horizontal and vertical stripes of equal width. Self-attention is then computed in parallel in both planes. Multi-head attention is divided into two groups, where self-attention is performed separately before being concatenated, which can be seen in Fig. 5. Unlike window-based methods, which apply identical self-attention across all heads, CSWin expands the attention area by employing a multi-head grouping strategy, improving the models representative capacity and resultantly the performance.

Pale Transformer (Wu et al., 2022a) restricted the attention computation to a pale-shaped region that contains interlaced rows and columns. Self-attention scores are computed within row and column token groups as demonstrated in 5. This helps to reduce the computational cost by focusing on nearby tokens while allowing limited attention to distant tokens, effectively balancing local and global context capture and achieving better performance.

In these methods, global interaction is approximated through sequential axis-aligned aggregation rather than simultaneous pairwise comparison. By factorizing 2D attention into two 1D passes, they maintain structured long-range communication while significantly reducing computational overhead. Compared to window-based attention, axial attention does not isolate tokens into rigid partitions, rather, it preserves connectivity across entire rows or columns. This allows faster propagation of information across large spatial extents with fewer stacked layers. However, interaction between off-axis tokens remains indirect, relying on composition across multiple passes, which may limit representational flexibility.

Geometric extensions such as DiagSWin (Li et al., 2024b) try to overcome one limitation of strict row–column decomposition by proposing diagonal-shaped windows to calculate attention scores within left and right diagonal-shaped windows, reducing computational cost while expanding visual attention. By altering the orientation of attention windows, it expands receptive geometry beyond orthogonal axes. Nevertheless, such modifications increase implementation complexity and provide incremental gains compared to the simplicity and hardware efficiency of window-based approaches.

Advantages and limitations

Axial self-attention occupies a distinct niche among efficient Transformers. Unlike window-based attention as in Section 3.4, which enforces spatial locality, axial factorization preserves global interactions along structured directions. This enables faster long-range propagation only in specific directions with the assumption that remaining directions will propagate through the attention computation.

Computationally, axial decomposition reduces quadratic complexity without approximations or token compression, making it appealing for

segmentation and restoration tasks that require spatial continuity. However, interactions are stronger along axes, while off-axis relationships require multi-step composition, increasing path length and limiting flexibility compared to full attention. When applied to very high-resolution or multi-spectral imagery, axial factorization can introduce aliasing-like artifacts due to small phase misalignments between row and column passes, effects not observed in full self-attention (Cao et al., 2025). Thus, axial attention is most effective when resolution is high and semantic structure is predominantly axial.

These drawbacks help explain why axial attention never dominated. Window-based Transformers 3.4 offer better hardware compatibility and integrate more easily with hierarchical pyramids, whereas axial designs need careful implementation for memory efficiency. Moreover, large-scale foundation models favor architectural uniformity and kernel optimization, which axis-wise decomposition complicates.

3.6. Token recalibration self-attention

The input image is typically divided into fixed patches in most VTs models, which are then processed using some variation of self-attention. This strategy enables global context modeling while being inefficient in the capture of fine-grained details. To solve this, recent papers have explored token manipulation techniques where tokens are modified to increase efficiency and accuracy. These methods target the quadratic cost of self-attention by altering the token set. This family of approaches is attractive because it can provide input-adaptive efficiency beyond fixed patterns. In this section, we delve into key research works that focus on token manipulation strategies. We divide these methods into two sub-categories. In Section 3.6.1, we consider Retained Standard Self-attention based techniques, and in Section 3.6.2, we discuss Modified Self-Attention.

3.6.1. Retained standard self-attention

The Data Efficient Image Transformer (DeiT) (Touvron et al., 2021a) model explores the distillation procedure in a ViT by using a distillation token which reproduces the label estimated by a teacher model. This allows the student DeiT Transformer to learn from the teacher's output. The model is trained with fewer parameters and in less time with small computational resources and achieves competitive results.

Token 2 Token (T2T) (Yuan et al., 2021c) progressively aggregates neighboring tokens into a single token, effectively modeling local structures such as edges and lines, which are crucial for image classification. This process reduces the length of the token sequence while embedding local context which helps the model to use self-attention to obtain both local and global information, reducing parameters and improving performance. DeiT and T2T show that token modification is not solely about pruning for speed, some approaches reshape tokenization to improve learning dynamics and locality. Distillation tokens enhance data efficiency and optimization, while progressive token aggregation introduces local structure priors before global attention. When token methods boost accuracy without heavy pruning, they typically improve locality and supervision rather than merely reducing token count.

QuadTree (Tang et al., 2022a) reduced the quadratic computational complexity of traditional VTs for dense prediction tasks by leveraging a coarse-to-fine approach. They construct a multi-level tokens pyramid by downsampling query, key and value and computes attention in the relevant regions determined by top-K patches, receptive field can be seen on 5. This allows the model to skip irrelevant regions and only concentrate on relevant parts.

Soft Token Pruning ViT (SPViT) (Kong et al., 2022) prunes less important tokens to reduce computational overhead while maintaining accuracy. It uses soft token pruning where the model learns to assign an importance score to tokens and drop redundant tokens. The key innovation is the integration of latency awareness into the pruning process which ensures real-world speed-ups rather than just FLOPs reduction.

The Adaptive Token Sampler (ATS) (Fayyaz et al., 2022) assigns significant scores to tokens based on the attention weights and selects the subset of tokens. As top-k selection does not adaptively select tokens and discards all lower-score tokens, ATS uses inverse transform sampling to sample tokens based on their score which helps to adjust the number of tokens processed based on input, leading to a compute reduction.

Evo-ViT (Xu et al., 2022) consists of two components, structure-preserving token selection and Slow-Fast token updating. The structure-preserving token selection differentiates between informative tokens and placeholder tokens by using global class attention. This attention calculates scores for these two token types and distinguishes them based on their score. Unlike other methods that directly drop uninformative tokens, Evo-ViT keeps tokens to maintain spatial structure. The slow-fast token updating mechanism updates the informative tokens through the slow path while updating the uninformative tokens using a representative token through the fast path. Its dynamic and structured token processing ensures that informative and uninformative tokens contribute to the final result, balancing efficiency and accuracy.

AdaViT (Meng et al., 2022) dynamically selects the patches, self-attention heads and Transformer blocks for each input image based on its characteristics by using a lightweight decision network. This network computes usage policy matrices to predict binary decisions indicating whether a patch should be preserved or discarded. This policy consists of three linear layers and sigmoid functions, utilizing the Gumbel-Softmax trick (Maddison et al., 2017) to enable differentiable sampling and improve the precision of decision-making.

DynamicViT (Rao et al., 2021) is proposed to remove redundant tokens using a prediction module in different layers. An attention-masking technique is used to remove tokens by blocking their interactions with other tokens. It combines local and global features and feeds them to an MLP block to predict the probability of keeping or dropping tokens in the prediction block. In attention masking, instead of removing any token, it stops token interaction, so the effects of non-informative tokens on informative tokens can be eliminated. This improves efficiency by gradually pruning 66% of the input tokens, with a minimal reduction in accuracy.

Patch Slimming (Tang et al., 2022b) involves the top-down pruning approach where the model eliminates the redundant patches layer-by-layer, starting from the final layer and then guides the process in the preceding layer. In each layer, patches are evaluated by calculating the importance score based on their impact on the final output, and those with lower scores are removed, reducing the model's cost.

Jiang et al. (2021b) introduce a new token labeling strategy where each patch is assigned an individual, supervised location-specific label generated by a machine annotator as opposed to applying a single label to an entire image. This approach encourages the model to learn fine-grained details, leading to improved performance.

Inspired by adaptive computation time (Graves, 2016), A-ViT (Yin et al., 2022) adjusts the number of tokens processed adaptively during inference using a halting module. A-ViT computes a halting probability for each token, allowing the model to focus on informative tokens and discard the less useful ones. It utilized the original network's parameters for learning the halting mechanism guided by distributional prior regularization to stabilize training.

Token pruning and sampling methods exploit image redundancy by making computation input-adaptive, typically via predictors that select which tokens or blocks to process. While aggressive pruning significantly reduces FLOPs, it can disrupt spatial locality and structural integrity. Structure-preserving variants, such as placeholder or representative tokens, aim to maintain geometric consistency. As a result, these methods often deliver strong gains in classification and latency, but require careful tuning and may generalize less reliably across domains and resolutions.

Long et al. (2023) and Expediting Vision Transformer (EViT) (Liang et al., 2022) incorporate both token importance and diversity for token

pruning, involving token decoupling and merging approaches. They first decouple the token sequence into attentive and inattentive tokens based on class token attention. Then, they cluster similar inattentive tokens and homogeneous attentive tokens while keeping the important attentive tokens using the cosine similarity score. This maximizes token diversity while preserving local information. Similarly, EViT focused on the attentiveness of class tokens by obtaining an attention score, indicating the importance of each token. Using this technique, most attentive tokens are preserved and less attentive tokens are fused into a token using a weighted average operation that helps reduce computational cost.

Token Merging (ToMe) (Bolya et al., 2023) also computes the similarity between each token using cosine similarity in two equally partitioned sets. The algorithm selects the most similar tokens via bipartite soft matching and merges them by averaging their features. Finally, these two sets are concatenated back together. This bipartite soft matching is nearly as fast as random token pruning and faster than k-means clustering (Lloyd, 1982), making it efficient for real-time applications.

Token merging and decoupling methods shorten sequences by aggregating redundant tokens instead of discarding them. Using strategies such as bipartite matching or cosine similarity, they fuse tokens to preserve denser representations than binary pruning. However, repeated weighted averaging can behave like a spatial low-pass filter, blurring boundaries and diminishing fine details. As a result, while merging improves optimization stability and throughput for classification, it may underperform on tasks requiring high-frequency precision, such as small-object detection or edge-sensitive segmentation.

3.6.2. Modified self-attention

These methods show that token modification is sometimes a proxy for improving training stability and inductive bias, not just reducing FLOPs. Chen et al. (2022a) claimed that the vanilla ViT is not optimized for many vision applications. They proposed RegionViT — a pyramid-based model that employs regional to local attention where the regional and local tokens are generated from an image with different patch sizes. The approach begins by dividing the input image into larger patches from which regional tokens are computed. Within each region, local tokens are computed using a smaller patch size. These tokens are then processed separately using a standard Transformer. The model first performs self-attention on regional tokens to capture global contextual information across different regions. After, local attention is performed by jointly attending to the local tokens within each region alongside their corresponding regional token. This approach allows regional tokens to pass global information to local tokens while simultaneously learning from them. In local attention, each regional token is combined with the local tokens in the same region, enhancing the overall feature representation. The model also employs a downsampling process to reduce spatial resolution by half while doubling the channel dimension, similar to CNN, for both regional and local tokens before moving to the next stage. At the end of the network, the remaining regional tokens are averaged to form the final embedding for classification, whereas for detection they utilize all local tokens at every stage. This approach significantly reduces memory consumption and improves performance.

A typical model usually inserts the class token in the Transformer block in the first stage, but Class Attention in Image Transformers (CaiT) (Touvron et al., 2021b) placed it in the middle stages. This helps the self-attention focus on input patches only at the beginning, helping the model learn patterns effectively without any interference from classification. By delaying the integration of the class token, the model avoids the issue of trying to learn the features and do the classification simultaneously, avoiding contradictory objectives and improving performance.

In the original self-attention, dropout is applied after the softmax operation which disturbs the probability distribution. DropKey (Li et al., 2023a) moved the dropout unit before the attention matrix calculation. Instead of using the key as attention weights, this approach utilizes the key as the dropout unit which helps to maintain regularization and probability distribution. The model starts with a higher dropout rate in starting layers and gradually decreases the rate across further layers, which avoids overfitting and captures more global information.

Lee et al. (2021) proposed two techniques, Shifted Patch Tokenization (SPT) and Locality Self-Attention (LSA), which help Vision Transformer (ViT) train on small-sized datasets from scratch. The high performance of ViT is largely attributed to its use of large datasets, and this dependence comes from low locality inductive bias. To improve the locality inductive bias of ViT, SPT increases the receptive field of visual tokens by spatially shifting the input image into four diagonal directions and concatenating these shifted images with the original. This process embeds more spatial information into the visual tokens, thereby enhancing the locality inductive bias of ViT. LSA uses diagonal masking to exclude self-token relations from the softmax operation and force the model to focus more on inter-token relations. It further incorporates learnable temperature scaling which allows the model to adjust the softmax temperature during training that can be seen in Fig. 4 as SCL (second from left). They sharpen the attention score distribution and makes the attention more focused, thereby enhancing the locality of the attention mechanism. These approaches show that strengthening inductive bias and regularization often matters more than merely reducing parameters. By redesigning tokenization or refining attention objectives, they alleviate the data-hungry nature of Transformers. Their effectiveness lies in improving learning dynamics, enabling better convergence on small datasets and finer pattern recognition, while adding far less architectural and hardware overhead than aggressive pruning methods.

To avoid information loss, Multi-Criteria Token Fusion (MCTF) (Lee et al., 2024) was proposed to capture the relationship between tokens based on similarity, informativeness and size. MCTF computes the attraction score between tokens based on similarity, informativeness and size. To reduce the amount of redundant information, tokens are eliminated using a cosine similarity score. To avoid information loss, an average attention score is measured to indicate the impact of each token on others. The size of the tokens indicates the number of fused tokens. Smaller tokens are fused to prevent the creation of large tokens that can lead to significant information loss. Further, it uses Bidirectional Bipartite Soft Matching to merge tokens by considering bidirectional relations between token sets, reducing the computational cost of similarity calculations. To improve the informativeness, one-step-ahead attention is used which considers the attention map of the next layer, helping in retaining more relevant tokens. MCTF achieves a significant reduction in computational complexity while slightly improving the accuracy.

This group of methods modifies the connectivity pattern of attention, such as PaCa (Grainger et al., 2023) which transforms patch-to-patch attention to patch-to-cluster attention by grouping patches into predefined clusters and computing the attention score between images patches (queries) and clusters (keys and values). This applies clustering before computing attention score which reduces the number of interactions between patches, saving computational resources. This technique is particularly beneficial for tasks involving high-resolution images.

Shunted self-attention (SSA) (Ren et al., 2022) selectively merges tokens to represent larger object features while preserving fine-grained details by maintaining some tokens through a multi-scale token aggregation (MTA). Different receptive field sizes are a assigned to different heads with the same layer, unlike vanilla attention which uses uniform sizes across all heads. This allows the model to capture both local and global information simultaneously and handle multi-scale objects, leading to improved performance.

BiFormer (Zhu et al., 2023) proposed Bi-level routing attention (BRA) in which each query attends a small number of key–value

patches instead of all. BRA filters out irrelevant patches in a coarse region using a region-level affinity graph to keep only the top tokens in each region and then it applies full attention to smaller fine-grained patches within the union of routed regions. This approach enables dynamic and query-aware sparsity, allowing each query to attend to a small subset of the most semantically relevant key–value pairs, thereby reducing computational complexity while maintaining high performance.

Sparsifiner (Wei et al., 2023) enhanced the efficiency through learning sparse, instance-dependent attention patterns which utilizes a lightweight connectivity predictor module that dynamically estimates the relevance between token pairs based on spatial or semantic features. This helps the model to adjust its attention based on the unique features of the input, which can vary from one image to another, rather than using fixed patterns, making it more flexible and scalable for large-scale vision tasks, as shown in Fig. 5.

Dynamic Group Attention (DGA) (Liu et al., 2022) divided all queries dynamically into multiple groups using a k-means clustering algorithm. From these, the most relevant keys and values for each group were selected by computing scores between them, as demonstrated in 5, reducing computational complexity.

Associative Transformer (AiT) (Sun et al., 2025) proposed a sparse, memory-augmented self-attention mechanism that improves efficiency and reasoning capability. Instead of attending to all token pairs, they use a top-k bottleneck attention guided by a set of learnable priors. These priors act as attractors in a continuous Hopfield network, enabling the model to reconstruct and enrich token representations after sparsified attention. A bottleneck balance loss ensures diverse token selection across layers, preventing redundancy. This design reduces computation, improves parameter efficiency, and delivers strong performance.

These methods improve efficiency by sparsifying the attention matrix rather than shortening the sequence. Methods such as region-level routing, patch-to-cluster bottlenecks, and learned connectivity predictors restrict each query to a semantically relevant subset of keys and values. Unlike pruning, these query-aware sparsity schemes preserve full spatial resolution, which is crucial for dense prediction tasks. However, their dynamic and often hard-to-optimize attention graphs make performance highly dependent on reliable routing. If long-range dependencies are missed early in training, global reasoning may remain permanently constrained.

Advantages and limitations

Token pruning, merging, reweighting, and routing reduce self-attention's quadratic cost by modifying the token set or its connectivity. Sequence reduction and interaction sparsification, techniques emerge in this section. Sequence reduction offers clear latency gains and supports high-resolution inference by reducing token count, while interaction sparsification preserves spatial detail by retaining tokens but limiting pairwise interactions. Unlike windowing (Section 3.4) or axial (Section 3.5) techniques, token-based methods can be input-adaptive, allocating compute to semantically salient regions, especially useful when images contain large redundant backgrounds.

The core limitation is information irreversibility. Once tokens are pruned or heavily merged, fine details and precise geometry may be lost, which can harm dense prediction and small object recognition. Many methods also rely on auxiliary predictors such as importance estimators or routing modules, adding extra hyperparameters and potential training instability, especially in early training when attention patterns are still forming. In addition, token dynamics can disrupt spatial structure. Uneven pruning may distort positional relationships, and repeated merging can behave like a smoothing operator that blurs boundaries. Furthermore, current methods often lack theoretical grounding on how token dynamics affect attention convergence and representational capacity, leaving open questions about optimal token management strategies in large-scale or real-time vision applications. Input-adaptive sparsity often creates irregular memory access patterns that may not achieve theoretical speedups on standard GPUs. Aggressive token dynamics can create distribution-dependent behavior that is hard to guarantee under transfer. As a result, token methods are most compelling when latency or resolution constraints dominate, while more uniform structural strategies such as hierarchical 3.2 or window-based 3.4 backbones remain preferable when robustness and general-purpose transfer are the main goals.

3.7. Reformulated self-attention

Reformulation-based attention methods modify the mathematical operator of self-attention rather than restricting tokens scope as in Sections 3.4 and 3.5, reorganizing resolution as in Section 3.2, or changing token counts as in Section 3.6. Their goal is to break the quadratic cost $O(n^2)$ of softmax attention through approximations or alternative similarity and normalization schemes such as kernelization, low-rank factorization, sparsification, or normalization replacement. However, altering the attention formula also changes optimization behavior and numerical stability, creating a trade-off between efficiency and faithful global interaction modeling. This section reviews representative reformulations and highlights patterns that explain their practical adoption.

Early reformulations often change how similarity is computed while preserving the overall softmax attention pipeline. Adder Attention (Shu et al., 2021) was proposed which includes a general adder linear transformation operation to replace the traditional multiplication version in both feed-forward modules and self-attention layers in the Transformer. Specifically, it replaces the standard scaled dot-product with l-the distance between the query and key to measure similarity in the self-attention module. This change introduces a low-rank property in the attention matrix, concentrating information on a few large singular values which can cause the functionality of attention not to preserve the data. To address this, an extra identity mapping was inserted in the adder attention module to homogenize the distribution of singular values and preserve the rank, reducing energy consumption while maintaining competitive performance.

$$Q_{m,j,q} = -\sum_{d=1}^{D} |X_{j,d} - W_{Q,m,q,d}|, \tag{2}$$

$$K_{m,i,k} = -\sum_{d=1}^{D} |X_{i,d} - W_{K,m,k,d}|, \tag{3}$$

$$V_{m,i,v} = -\sum_{d=1}^{D} |X_{i,d} - W_{V,m,v,d}|. \tag{4}$$

Eqs. (2)–(4) represent the adder linear transformation while Eq. (5) is Adder MHSA.

$$A_{m,j,i} = \frac{\exp\left(-\frac{\|Q_{m,j,:} - K_{m,i,:}\|_1}{\sqrt{d_a}}\right)}{\sum_{i=1}^{N_{kv}} \exp\left(-\frac{\|Q_{m,j,:} - K_{m,i,:}\|_1}{\sqrt{d_a}}\right)}, \tag{5}$$

where $d_a = 2d_t\left(1 - \frac{2}{\pi}\right)$ and $d_t = d_h$.

DeepViT (Zhou et al., 2021a) observes that the performance of VTs saturates as the model depth increases due to the attention collapse issue. To address this, Re-Attention was proposed as in Eq. (6), which re-generates attention maps by mixing information from different heads, thereby reducing the similarity between attention maps across layers.

$$\text{Re-Attention}(Q, K, V) = \text{Norm}\left(\Theta^{\top}\left(\text{Softmax}\left(\frac{QK^{\top}}{\sqrt{d}}\right)\right)\right)V \tag{6}$$

where Θ is the learnable transformation matrix and Norm is a normalization function used to reduce layer-wise variance.

These approaches emphasize structural stability and efficiency without fundamentally altering the architecture. By retaining the softmax-based framework, they maintain stable gradients while targeting issues such as attention collapse and head redundancy or hardware energy use. Instead of merely lowering complexity, they refine attention's internal structure, improving rank and cross-head diversity, to prevent information loss or performance saturation as models scale in depth or face hardware constraints.

Deformable Attention (DAT) (Xia et al., 2022) was introduced to model the relations among tokens under the guidance of important regions in feature maps. These important regions are selected by deformed sampling points which are learned from the queries by an offset network. DAT uses bilinear interpolation to sample features which are fed to Key and Value projections to get deformed Keys and Values. The standard MHSA is then applied to attend queries to sampled keys and aggregate features from deformed values as shown in Fig. 5. Additionally, the location of deformed points provides better relative position bias to facilitate the learning of deformable attention. The model is based on Swin architecture and it uses the deformable attention replacing MHSA but only at the last two stages. The authors argue that Swin is ineffective for capturing the global receptive field or handling large objects in inputs. Therefore, they prioritize global receptive field modeling while utilizing Swin for local attention, demonstrating competitive results.

The output of an attention head is formulated as:

$$z^{(m)} = \sigma\left(\frac{q^{(m)}\tilde{k}^{(m)\top}}{\sqrt{d}} + \phi(\hat{B}; R)\right)\tilde{v}^{(m)}, \tag{7}$$

where $\phi(\hat{B}; R) \in \mathbb{R}^{HW \times H_G W_G}$ corresponds to the position embedding, and σ denotes the softmax function.

ViTALiTy (Dass et al., 2023) involves a combination of low-rank and sparse approximations to reduce complexity through a linear Taylor attention which approximates the softmax attention using a first-order Taylor series expansion. The key idea is the decoupling of softmax attention into weak and strong attention where the weak component is calculated using the linear Taylor attention and the strong component is approximated using sparsity.

KVT (Wang et al., 2022a) proposed k-nearest neighbor attention that selects the top-k most similar tokens for each query to compute the attention map, rather than using all tokens as in self-attention. It releases two versions, slow and fast K-NN, which use Euclidean distance and dot product, respectively. Attention is calculated against all the keys and the top-k elements are selected. This helps filter out irrelevant or noisy tokens, improving performance.

$$A_i = \text{softmax}\left(\frac{\langle q_i, (k_{j_1}, \ldots, k_{j_l}, \ldots, k_{j_k})\rangle}{\sqrt{d}}\right), \quad k_{j_l} \in N_i^k \tag{8}$$

where q_i is the ith query, k_{j_l} are the keys corresponding to the top-k nearest neighbors of q_i, N_i^k is the set of indices of the top-k nearest neighbors of q_i, and d is the dimension of the keys and queries.

Flatten ViT (Han et al., 2023) proposed focused linear attention that addresses the inability of self-attention to focus on important features by using a focused function ϕ_p. They adjust the feature direction of queries and keys to achieve a distinguished difference between them. To maintain feature diversity, depthwise convolution is used in the attention matrix which helps the model to focus only on adjacent features in space.

Zhou et al. (2023) investigated the performance of local self-attention (LSA) and its counterparts, such as convolution and dynamic filters, to find out their performance and limitations. They analyzed them using different channel settings and spatial processing and identified that the relative position embedding and neighboring filter application are the key factors affecting performance. Based on these findings, an Enhanced LSA (ELSA) was proposed that changes the attention with the Hadamard product replacing the dot product between queries and keys to efficiently generate attention in the neighboring area as seen in Eq. (9), where $\odot$ is the Hadamard product. They also use a ghost head inspired by GhostNet, which combines attention maps with static matrices to increase channel capacity. The ghost head module expands the original heads by combining Hadamard attention with two static filters. Experiments show that, without other architecture/hyperparameter modifications, ELSA can consistently improve the baseline, regardless of the model size and tasks, with little overhead being introduced.

$$h_{j-i} = \text{Softmax}_j\left((q_i \odot k_i)r^k_{j-i} + r^q_{j-i}(q_j \odot k_j) + r^b_{j-i}\right) \tag{9}$$

LeViT (Graham et al., 2021) created a multi-stage Transformer architecture with a pyramid structure similar to CNNs. It removes the classification token and replaces it with average pooling on the last feature map. LeViT further introduces several innovations related to attention mechanisms. They introduce a learnable attention bias as in Eq. (10) by replacing the traditional positional embedding and adding a bias term to the attention maps. This allows each head to learn separate biases and improve the model's ability to capture spatial relationships. They also employ a computationally efficient patch descriptor and a redesigned Attention-MLP block to improve network capacity. Finally they also use a shrinking attention block for downsampling to reduce the spatial resolution of activation maps while increasing the number of channels, similar to convolutional layers. LeViT is able to perform efficiently due to attention bias that reduces computational complexity while preserving accuracy.

$$A^h_{(x,y),(x',y')} = Q_{(x,y)}, : \cdot K_{(x',y')}, : + B^h_{|x-x'|,|y-y'|} \tag{10}$$

where $B^h_{|x-x'|,|y-y'|}$ is the attention bias for head h.

Learned sampling, similarity-based pruning, or localized biases reduce computational overhead by sparsifying the attention matrix. By restricting the key–value pairs available to each query, these methods preserve the sharp activations of softmax attention and often outperform linear kernels in capturing fine details. Depending on the selection strategy, DAT and KVT aim to retain global context through informative token retrieval, whereas local-focused models like ELSA favor spatial inductive biases over full-range interactions. The main risk is irreversible loss of long-range dependencies if sampling or window design is misaligned with task demands, leading to inconsistent gains across datasets.

SOFT model (Lu et al., 2021) is a softmax-free self-attention mechanism that removes the softmax function and replaces the dot-product with a Gaussian kernel function without any further normalization, as can be seen in Eq. (11). The computed attention matrix is symmetric and all diagonal elements hold the largest values, with the off-diagonal elements being close to zero, helping complexity to be linear. To improve convergence, a low-rank matrix decomposition is used to approximate the attention by calculating its Moore–Penrose inverse (Penrose, 1955) using the Nystrom (Williams and Seeger, 2000) approximation method. This approach enables the model to handle much longer sequences and provides better performance.

$$S_{i,j} = \exp\left(-\frac{1}{2\sqrt{d_e}} \cdot \|Q_{i,:} - K_{j,:}\|_2^2\right) \tag{11}$$

Cross-covariance attention (XCiT) (Ali et al., 2021) is a self-attention mechanism that operates on feature channels instead of token dimensions. Attention scores are computed based on a cross-covariance matrix of normalized query and key where normalization is used to stabilize training with a learnable parameter to adjust the distribution of attention weights. This attention is further optimized by dividing features into heads to control feature interactions within tokens to handle higher dimensional data, thus reducing complexity.

$$XC - Attention(Q, K, V) = Softmax\left(\frac{\hat{K}^\top \hat{Q}}{\tau}\right) V \tag{12}$$

where matrices $\hat{Q}$ and $\hat{K}$ are the normalized versions of Q and K using ℓ_2-normalization across the token dimension. The parameter τ is a

learnable temperature parameter that scales the inner products before applying the softmax function.

Sinkformers (Sander et al., 2022) used Sinkhorn's algorithm (Sinkhorn, 1964) to make attention matrices doubly stochastic instead of softmax which normalized it to row-wise stochastic. The authors show that this change allows for a better theoretical understanding of attention mechanisms, linking them to Wasserstein gradient flows and diffusion processes.

SimA (Koohpayegani and Pirsiavash, 2024) attention simply normalizes query and key matrices with ℓ_1-norm before multiplying them. It allows the attention mechanism to be simplified to a straightforward matrix multiplication without the need for the softmax operation, reducing computational complexity and maintaining accuracy, as in Eq. (13), where $\|\cdot\|_1$ denotes the ℓ_1-norm.

$$\hat{Q}_i = \frac{Q_i}{\|Q_i\|_1}, \quad \hat{K}_i = \frac{K_i}{\|K_i\|_1} \tag{13}$$

Mao et al. (2022) replaces the scaled dot-product with a learnable position importance matrix, revealing the importance of each query–key pair. This helps to filter out the noisy correlations and strengthens the model's robustness. It can be seen in (14), where $\odot$ denotes the element-wise product.

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top \odot W_p}{\sqrt{d}}\right)V, \tag{14}$$

Castling-ViT (You et al., 2023) introduces linear-angular attention which substitutes dot-product with angular kernels to measure similarities between queries and keys using spectral angles. It decomposes these kernels into linear terms and high-order residuals, keeping only the linear terms for inference to reduce complexity. The framework further employs depthwise convolution and auxiliary masked softmax attention to approximate the high-order residuals. The masks for softmax attention are regularized to gradually become zero, ensuring no additional inference overhead.

$$\begin{aligned} H &= \text{Sim}(Q, K) \cdot V \approx \frac{1}{2} \cdot V + \frac{1}{\pi} \cdot Q \cdot (K^\top \cdot V) \\ &+ M_{\text{DW}} \cdot V + M_{\text{SparseAttn}} \end{aligned} \tag{15}$$

where $M_{\text{DW}} \cdot V$ represents the matrix form of the depthwise convolution applied to the value tokens, and $M_{\text{SparseAttn}}$ denotes the normalized sparse softmax attention.

Vicinity ViT (Sun et al., 2023) takes inspiration from cosFormer (Qin et al., 2022) and replaces softmax with the Rectified Linear Unit (ReLU) layer and adjusts attention weights based on 2D Manhattan distance between image patches. It further computes attention in a compressed feature space which gives linear complexity and uses a skip connection to capture global features.

Eqs. (16), (17), and (18) show the main operations from Sun et al. (2023):

$$\begin{aligned} Q'_i &= [\phi^{\cos a_i}(Q_i), \phi^{\sin a_i}(Q_i), \phi^{\cos b_i}(Q_i), \phi^{\sin b_i}(Q_i)] \\ K'_j &= [\phi^{\cos a_j}(K_j), \phi^{\sin a_j}(K_j), \phi^{\cos b_j}(K_j), \phi^{\sin b_j}(K_j)] \end{aligned} \tag{16}$$

$$\begin{aligned} \phi^{\cos y}(x) &\triangleq \cos(y)\phi(x) \\ \phi^{\sin y}(x) &\triangleq \sin(y)\phi(x) \end{aligned} \tag{17}$$

$$O_i = \frac{Q'_i \left(\sum_{j=1}^{N} (K'_j)^\top V_j\right)}{Q'_i \left(\sum_{j=1}^{N} (K'_j)^\top\right)} \tag{18}$$

Eq. (16) defines the transformed query Q'_i and key K'_j vectors after applying the cosine and sine functions with the positional encodings a_i, a_j, b_i, b_j. Eq. (17) provides the definitions of the cosine and sine re-weighting functions $\phi^{\cos y}(x)$ and $\phi^{\sin y}(x)$. Eq. (18) shows the final output O_i of the attention mechanism using the transformed queries and keys.

Linearized self-attention (Fan et al., 2025) also replaces the softmax operation in standard self-attention with a kernel function $\phi(.)$ that enables computing attention in O(n· d) time instead of O(n^2), by reordering matrix multiplications. A kernel feature map is used for queries and keys and leverages linear attention for efficiency.

Kernelized and softmax-free attention offers attractive $O(n)$ scaling, but the empirical gap to softmax attention frequently persists because softmax provides a strong inductive bias, producing sharp, competitive distributions that naturally emphasize salient tokens. In contrast, many linearized approximations suffer from a low-rank dilemma, behaving like low-pass filters that average token information and lose fine-grained detail. To bridge this gap, methods incorporate auxiliary mechanisms, such as Nyström approximations, depthwise convolutions, or complex positional re-weighting. Paradoxically, as these approximations become more effective at mimicking the discriminative power of softmax, they often reintroduce the computational overhead or architectural complexity they originally sought to eliminate.

VOLO (Yuan et al., 2022) introduced Outlook attention where input tokens are dynamically split into smaller patches and compute similarity with their surrounding neighbors using a reshaping operation instead of matrix multiplication. Then it adopts a sliding window mechanism to locally encode token representations at a fine level which are further fed into subsequent layers of the self-attention to extract global information at the coarse level, leading to improved performance.

Wave-ViT (Yao et al., 2022) incorporates wavelet transforms to enable lossless down-sampling of the input patches before performing self-attention. Discrete Wavelet Transform (DWT) is used to decompose the input into different frequency subbands, allowing for efficient down-sampling while preserving the original information. Additionally, the inverse wavelet transform (IDWT) is applied to enhance the self-attention outputs by aggregating local contexts with an enlarged receptive field, enhancing performance.

MiniViT (Zhang et al., 2022c) proposes Weight Multiplexing which consists of weight sharing, transformation and distillation techniques. It shares the weights across Transformer blocks and increases the parameter diversity and stability via transformations. Additionally, the distillation transfers the weight knowledge from large to small models, making the model more efficient.

Some models treat attention as a signal-processing operator, explicitly correcting low-pass behavior, such as Wang et al. (2022b), investigated the performance degradation in VTs using Fourier spectrum domain analysis. They show that the self-attention acts as a low-pass filter, causing loss of high-frequency data in deeper layers, leading to attention collapse and patch uniformity. To address these issues, they propose attention scaling (AttnScale) and feature scaling (FeatScale). AttnScale decomposes the attention matrix into a low-pass filter and a high-pass filter, then rescales them to create an all-pass self-attention matrix. FeatScale re-weights feature maps across different frequency bands to amplify high-frequency signals, improving the low-pass filtering effect. Both components allow ViT to go deeper without significant performance loss.

Instead of fully linearizing attention, these methods emphasize signal fidelity and representational richness. They view attention as a signal-processing operator and counter its natural low-pass behavior, which can cause patch uniformity and suppress high-frequency details in deep models. By preserving spectral diversity in the information flow, they show that true efficiency is not just about lowering asymptotic complexity, but about sustaining fine-grained discrimination as depth increases.

Advantages and limitations

Reformulating self-attention can reduce computational and memory costs, enabling scaling to higher-resolution inputs or longer sequences without prohibitive resource use. Methods like sparse factorizations (Dass et al., 2023), low-rank approximations (Lu et al., 2021), and kernelized attention (You et al., 2023) allow VTs to handle high-resolution inputs or dense feature maps with manageable memory and computation. However, these methods can introduce additional

hyperparameters and architectural complexity, making model tuning more challenging and reducing the interpretability of learned attention patterns. In practice, the main limitation is that softmax attention is difficult to approximate without losing its competitive winner-take-more behavior. Fan et al. (2025) shows that linear attention often exhibits an accuracy gap because approximations implicitly smooth attention distributions, weakening long-range selectivity. Attempts to close this gap, such as those of Lu et al. (2021) and Sander et al. (2022), achieve improvements but still fall short of matching the performance of their softmax-based counterparts. Nyström variants (Lu et al., 2021) or other compression (Sun et al., 2023) can assign less total weight than they should to some tokens, effectively underestimating the importance and leading to dropped long-range interactions. Many reformulations introduce non-standard operations, approximation hyperparameters, or training instabilities that complicate scaling and hardware optimization. Consequently, large models more often adopt structurally efficient designs such in Section 3.2, that preserve standard softmax attention while improving scalability through architectural constraints while reformulated attention remains valuable when specific deployment constraints dominate. Moreover, large foundation models benefit significantly from hardware-optimized dense matrix operations, whereas alternative formulations often lack comparable kernel-level optimization, reducing real-world efficiency gains.

3.8. Attention fusion

Attention fusion approaches combine multiple attention paradigms within a single block or pipeline, such as local and global or spatial and channel, to capture complementary inductive biases. Fusion can be seen as an orthogonal design axis to the structural, such in Sections 3.2 and 3.4 and operator-level 3.7 strategies. Rather than changing where attention operates, this category changes how different attention mechanisms cooperate. In this section, we highlight key works that leverage such fusion, exploring how these hybrid strategies push the boundaries of performance and efficiency in ViT architectures.

Stacking local and global attention in the same block is a pragmatic route to capture both fine detail and long-range context without abandoning the standard softmax kernel. Keeping in mind the lack of scalability of the self-attention mechanism, Tu et al. (2022) introduced an efficient and scalable multi-axis attention model (MaxViT) which includes blocked local attention and grid global attention. To capture local attention, it uses a shape of $(\frac{H}{P}\text{x}\frac{W}{P}, P\text{x}P, C)$, while to capture global attention it uses $(G\text{x}G, \frac{H}{G}\text{x}\frac{W}{G}, C)$, where HW is spatial dimension and P, G are patch size from both attention where P=G. These two attention mechanisms are stacked in a single block. It achieves state-of-the-art performance on a variety of vision tasks and scales extremely well to massive-scale data sizes. It demonstrates that structured factorization of spatial axes can approximate global receptive fields without computing full attention matrices.

Zhu et al. (2021) combines short-term attention and long-range attention via a segment-wise sliding window and dynamic low-rank projections, respectively. This technique helps to capture both local and global information from the image. In short-term attention, the input is divided into disjoint segments, where all tokens within its home segment and neighboring tokens are attended to gather fine-grained local correlations. On the other side, long-range attention is parametrized using dynamic low-rank projection where a learnable projection matrix reduces the dimensionality of key and value embeddings. This helps each query to attend to all the tokens without much complexity. Later, both attention are combined by allowing every query to integrate the information of keys and values from local attention and global attention. This combined approach makes it possible to achieve efficiency and effectiveness in capturing global and local dependencies. While MaxViT factorizes spatial axes, it factorizes the attention rank, reducing complexity while preserving complementary context scales.

ScalableViT (Yang et al., 2022b) proposed Scalable Self-Attention (SCSA) and Interactive Window-based Self-attention (IWSA). SCSA scales spatial and channel dimensions of query, key, and value to control complexity while preserving context awareness. IWSA reconnects non-overlapping windows by remerging value tokens and integrating neighboring spatial information to recover long-range dependencies. A dynamic positional encoding generator links the two modules. Together, they reflect a common pattern of efficiency-driven reformulation followed by interaction recovery.

These methods follow a shared principle of preserving efficient local computation while introducing structured global connectivity without full vanilla MHSA. They differ mainly in how global context is approximated, whether through grid partitioning, low rank projection, or window interaction mechanisms. Together, they show that modeling long-range dependencies does not require a full attention matrix when spatial or rank structure is efficiently utilized.

More methods decompose attention across spatial and channel dimensions, treating them as complementary axes. Cascaded group attention (Liu et al., 2023b) feeds each attention head with different splits of the full feature set, decomposing the attention computation across heads. This approach promotes attention diversity and reduces computational redundancy.

Dual Attention (DaViT) (Ding et al., 2022) is a technique where the model leverages both spatial window attention and channel group attention in one block to capture local and global information. Spatial window attention computes self-attention within local windows and is confined to smaller regions to reduce the computational complexity. Channel attention, on the other hand, operates on the transpose of the token matrix, treating channels as tokens to capture global information with linear complexity. DaViT is further optimized by grouping channels and applying self-attention within each group which ensures global interactions. These attention mechanisms complement each other, combining local and global details efficiently and handling high-resolution inputs effectively.

CoAtFormer (Chang et al., 2024) introduced a composite attention module that contains two branches. The first branch uses a dynamic channel module for global channel interactions through a graph neural network. The second branch, a spatial attention module, replaces self-attention and focuses on spatial global context. They compute the attention score with linear complexity using a learnable global token vector.

LightViT (Huang et al., 2022) involves aggregated self-attention with local–global broadcast and aggregated feed-forward networks with bi-dimensional attention. The aggregated self-attention uses learnable global tokens to capture global information and broadcasts them to local tokens which helps the model to gather global context and enhance local features. The bi-directional attention captures spatial and channel dependencies using channel attention and spatial attention branches. The channel attention branch captures global representations by averaging input features on spatial dimension and applying a linear transformation. The spatial attention branch models pixel-wise relationships by concatenating global and local token features, enabling the model to focus on salient tokens and better capture the image structure.

These methods treat spatial correlation and channel dependency as complementary information axes. By factorizing attention, they reduce the redundancy of vanilla MHSA either through alternating spatial and channel blocks or parallel branches. This explicit separation, followed by targeted fusion, preserves fine-grained local detail while modeling global semantics at significantly lower computational cost.

Further models show that uniform attention over all tokens is inefficient and unnecessary. They introduces selective computation or routing mechanisms to fuse attention more sparsely. Partial Attention (Vo et al., 2024) uses a technique of choosing top-scoring tokens using mean for the relevant part, considering it as foreground and less scoring tokens as background. The background is computed using efficient single-query attention (SQA) where a learnable query token extracts

local information from the background while the foreground, containing relevant tokens, is calculated using mixed multi-head self-attention (MMSA) to capture important features.

MAFormer (Sun et al., 2024a) presented Multi-scale attention fusion which explores local aggregation and global feature extraction using local-window attention and global learning downsampling modules, respectively. It then combines global and local modules using an attention fusion module which captures the correlation between each local–global token.

DeBiFormer (BaoLong et al., 2024) addresses the limitations of semantic relevance in key–value pairs by deformable points in DAT (Xia et al., 2022) and the influence of irrelevant attention calculation of queries in Biformer (Zhu et al., 2023). It introduces a new module that combines the deformable attention from DAT with bi-level routing attention from Biformer as in 5. DeBiFormer first locates deformable points and then aggregates information from relevant key–value pairs. Uniform attention is often redundant and can be replaced with content-aware routing and selective fusion. By prioritizing foreground tokens or dynamically sampling deformable key–value pairs, computation is focused on the most informative regions. This prunes irrelevant interactions while preserving a sparse yet powerful capacity for global modeling.

Finally, some works extend fusion beyond individual blocks to the network level. Broad attention (BViT) (Li et al., 2023d) explored the transmission and integration of information from different layers using the broad connection where each layer can access and incorporate the attention maps or embeddings from all other layers. The output is then fused using parameter-free attention via weighted summation and average pooling which extracts the relationship between them, enhancing the model's ability to focus on critical aspects of the input.

Attention mixing (Zhou et al., 2022a) increases the number of attention maps via expansion and mixing while maintaining embedding dimensions. They further applies cross-layer regularization to increase the diversity of attention maps and utilizes convolution to enhance local token interactions which improves token aggregation and performance in the model. These methods shift the focus from intra-block efficiency to network-wide synergy. By sharing attention maps across layers and enabling expansive mixing, these approaches alleviate the information bottleneck of strictly hierarchical designs. Unlike block-level fusion strategies like DaViT (Ding et al., 2022) spatial–channel interleaving, such global interactions promote better feature reuse and help preserve fine-grained geometric details in deeper, more abstract layers.

Advantages and limitations

Fusion improves dense task accuracy and robustness to scale variation by combining complementary inductive biases that a single attention operator cannot provide. By integrating multiple attention types across scales, it strengthens feature representation, captures both local detail and global context, and models complex cross-region interactions. This often enhances accuracy and efficiency without requiring deeper models.

Fusion of multiple attention mechanisms adds computational complexity and increases memory demands, often increasing the already high resource consumption typical of ViTs. Additionally, the increased architectural complexity can complicate model training, requiring careful hyperparameter tuning and risk of overfitting, particularly when data is limited. The mechanisms struggle with maintaining an effective balance between preserving hierarchical feature relationships and capturing comprehensive global context. When two operators capture highly overlapping information, the marginal gain from fusion can be small while resource cost remains high. There is no one solution for which attention types should be fused, when fusion should occur, or how to balance representational capacity against redundancy. These open questions form clear future paths for research. Fusion is most powerful when task specificity or high resolution demands justify added complexity, such as in dense prediction and high resolution settings. It is less attractive as a default backbone for foundation models that prioritize simplicity, kernel efficiency, and broad transferability. Fusion should be read as a compositional axis, it answers the "how do attention primitives cooperate?" question, complementing the where 3.2, 3.4 and the what 3.7 axes discussed in earlier sections.

Future work should focus on principled metrics to measure complementarity, lightweight fusion primitives, and automated design strategies, such as architecture search for selecting fusion points, enabling the benefits of fusion without its full cost.

4. Self-attention in vision foundation models

Early research on vision Transformers primarily focused on mitigating the computational and memory bottlenecks of self-attention, as discussed in Section 3. Recently, the research focus has shifted toward data-centric scaling and large-scale pretraining, giving rise to vision foundation models. In this paradigm, self-attention is no longer optimized solely for efficiency on small or medium-scale datasets, but rather evaluated by its ability to support representation learning across massive datasets, diverse objectives, and multiple downstream tasks.

Self-attention plays a central role in vision foundation models due to its flexibility and expressive capacity. Unlike convolutional operators, self-attention's ability to aggregate global context from the earliest layers and long-range dependency modeling makes it suitable for large-scale pretraining. For example, SAM (Kirillov et al., 2023) leverages Transformer-based encoders with self-attention to produce task-agnostic image embeddings that generalize across segmentation prompts. Although SAM includes task-specific components, its backbone relies on scalable self-attention to enable zero-shot generalization. The image encoder computes attention once per image and reuses the resulting embedding across prompts, while the decoder applies cross-attention, with prompt queries attending to dense image keys and values.

Florence (Yuan et al., 2021b) uses a hierarchical ViT as its image encoder paired with a text Transformer, where self-attention is computed within local windows in a multi-scale hierarchy rather than globally over all patches as in the original ViT. It is trained on a Unified Contrastive learning objective where the model integrates specialized adapters to transition from coarse scene-level understanding to fine-grained object localization. It supports strong zero-shot, few-shot, and transfer performance across classification, detection, VQA, and video tasks. Florence-2 (Xiao et al., 2024) is an evolution of the original Florence that shifts from a two-tower contrastive framework to a unified, prompt-based, sequence-to-sequence architecture capable of handling a wide variety of vision and vision–language tasks with a single set of weights. Instead of training separate heads for each task, Florence-2 takes an input image plus a text prompt indicating the task, e.g., detect objects, and generates appropriate outputs in a unified way. Its core architecture combines a DaViT (Ding et al., 2022) encoder and a standard multimodal encoder–decoder Transformer that processes both visual and textual tokens together in sequence.

DINOv2 (Oquab et al., 2024) employs a standard ViT backbone trained via self-supervised learning on massive, curated image datasets. Despite minimal architectural modifications, the model achieves strong performance across dense prediction, image retrieval, and semantic understanding tasks. The difference in its attention mechanism is in how attention is trained and constrained. An automatic pipeline is built to curate a dataset of 142 million high-quality images (LVD-142M) from an uncurated crawl of 1.2 billion images. Training a 1-billion-parameter ViT with self-supervision is very unstable. Thus, LayerScale and a high degree of stochastic depth are utilized alongside the KoLeo regularizer to prevent feature collapse. FlashAttention is used to make training 2x faster with 3x less memory.

CLIP (Radford et al., 2021) optimizes ViT's self-attention to align image embeddings with text embeddings in a shared latent space using

a contrastive loss. Consequently, image self-attention prioritizes linguistically describable, semantically meaningful objects and attributes over purely visual or background features. Attention therefore learns to aggressively route information from relevant patch tokens into the text-based CLS token. SigLIP (Zhai et al., 2023) replaces the global softmax-based loss with a pairwise sigmoid loss. This decouples the batch size from the normalization constant, allowing for significantly more memory-efficient training at extreme scales and enabling better performance on smaller batch sizes compared to standard CLIP. Its loss gives direct gradients for each image–text pair, encouraging self-attention to capture subtle attributes (colors, materials, actions) more consistently. SigLIP allows attention to spread across multiple relevant regions, which improves multi-object and attribute understanding.

AIM (El-Nouby et al., 2024) orders image patches in a fixed raster scan and trains the model to predict the next patch given all previous ones. The goal is to scale vision models in a way that closely parallels LLM scaling, showing that larger models and more training data yield stronger visual features and downstream performance. It uses causal self-attention where each patch can only attend to preceding patches in the fixed order. This mimics next token prediction, where future tokens are unknown which gives AIM emergent sequence modeling capabilities for images, something standard ViT does not capture. AIMv2 (Fini et al., 2025) extends the original AIM framework by training a vision encoder using an autoregressive objective rather than standard masked-image modeling. Instead of reconstructing masked patches or relying purely on contrastive learning, AIMv2 treats visual tokens similarly to language tokens and predicts the next token in sequence using an autoregressive Transformer setup. It constrains the self-attention mechanism within the vision encoder by applying a prefix attention mask, which facilitates the use of bidirectional attention during inference without additional tuning. Compared to AIM, which focused on autoregressive image modeling for strong visual representations, AIMv2 scales the model and extends it to multimodal pretraining.

Detic (Zhou et al., 2022b) is a method to train object detectors with image-level labels. It decouples classification from localization so that large-scale classification data can be used to train the classifier part of a detector and improve detection performance. It uses Swin (Liu et al., 2021) and DETR (Carion et al., 2020) as backbones.

Masked Autoencoder (He et al., 2022) trains a ViT in a self-supervised way by randomly masking a high percentage of image patches and learning to reconstruct the missing patches from the visible ones using an asymmetric encoder–decoder architecture. Self-attention layers operate on a much shorter token sequence than in standard ViT, reducing compute and forcing the model to learn rich contextual representations from limited observations. A lightweight decoder takes the encoder outputs with learned mask tokens and applies standard MHSA over the full set of tokens to reconstruct pixel values. The encoder attends only to visible patches and the decoder attends to both real embeddings and mask tokens during reconstruction, which encourages learning of global structure from partial context.

Advantages and limitations

Vision Foundation Models represent a paradigm shift due to their exceptional generalization and zero-shot transfer capabilities. Unlike traditional models trained for a narrow task, they leverage self-attention to learn universal visual representations from massive, diverse datasets. This results in emergent properties, such as the ability to perform semantic segmentation or depth estimation without explicit supervision on those tasks. These models also reduce the need for extensive task-specific architectural engineering, as a single pretrained backbone can be adapted to classification, detection, segmentation, and retrieval tasks. Furthermore, they drastically reduce the need for labeled data in downstream applications.

However, these models carry significant limitations, primarily regarding computational, memory, data, and environmental impact. Training such models often requires specialized hardware and curated datasets, raising concerns about energy consumption, environmental impact, and data bias. In addition, the reliance on large-scale pretraining can obscure the contribution of architectural design choices, making it difficult to interpret model behavior or attribute performance gains. These limitations highlight the need for more efficient self-attention mechanisms, better understanding of scaling behavior, and principled approaches that balance architectural innovation with data-centric learning. For detailed overviews, see recent surveys on foundation models (Awais et al., 2025).

5. Post-ViT models

Despite the success of architectural optimizations such as window-based attention, hierarchical designs, and token pruning, they do not eliminate the computational and memory cost of self-attention drastically. As vision models increasingly scale to higher resolutions and longer sequences, these limitations motivate alternative sequence modeling paradigms with more favorable scaling properties.

State Space Models (SSMs) have recently emerged, offering linear-time complexity while retaining the ability to model long-range dependencies. Modern SSMs like Mamba (Gu and Dao, 2024) introduce selective state transitions and efficient parallel scan algorithms, enabling scalable training and inference. Unlike self-attention, which requires pairwise token comparisons with quadratic complexity, Mamba compresses sequence information into a structured hidden state using selective state updates, improving efficiency for long sequences. This section provides a technical exposition of how SSMs operate on visual data, emphasizing the architectural innovations that enable their competitive performance.

The Structured State Space Sequence model (S4) introduced efficient diagonal-plus-low-rank parameterizations but its parameters remained input-independent. The Mamba architecture (Gu and Dao, 2024) introduced the selective scan mechanism (S6), which enables the model to select which tokens to remember or forget, analogous to the ability of attention to focus on salient regions, addressing the code limitation of early SSMs. In vision tasks, selectivity allows the model (Liu et al., 2024b) to suppress background clutter and emphasize foreground objects, much like the query–key dot product in attention. These vision models (Zhu et al., 2024a) also address the one-dimensional limitation of SSMs through multi-directional or cross-scan strategies that enable comprehensive spatial interaction while preserving the linear computational complexity.

Fig. 6 illustrates different information propagation paradigms. Unlike convolution and window-based attention that model local interactions, and self-attention that relies on quadratic global aggregation, SSM-based methods capture long-range dependencies through structured directional scanning, enabling global context modeling with linear complexity. Notably, these solutions have resemblance to the axial and windowed attention mechanisms taxonomized in Sections 3.4 and 3.5 of this survey.

These SSM models can be divided into two sections: Pure SSMs and Hybrid Mamba-Transformer architectures that interleave both paradigms.

5.1. Pure SSM

Pure models rely entirely on state space layers, rather than self-attention or convolutions. They replace attention layers with Mamba blocks. In these models, an image is first split into patches and treated as a sequence using a state space mechanism which processes the sequence using learned dynamic state transitions that propagate information across tokens in a recurrent but parallelizable way. Here are a few Pure SSMs models.

VMamba (Liu et al., 2024b) also replaces quadratic self-attention with a linear-time mechanism. It stacks Visual State-Space blocks built around a novel 2D Selective Scan module, which also extends 1D

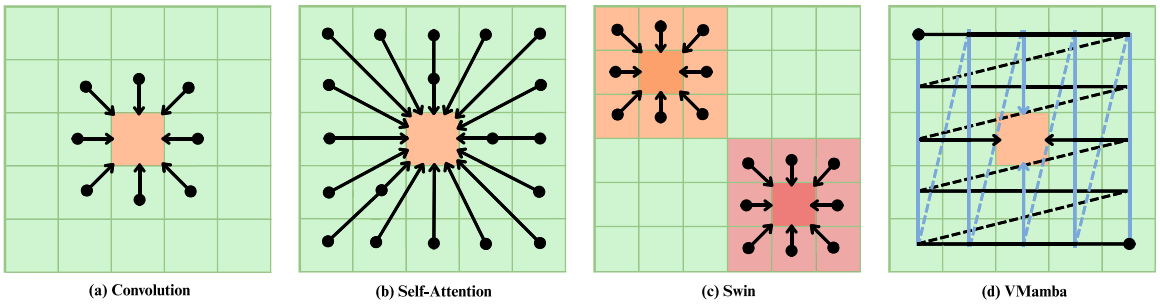


Fig. 6. Comparison of information flow patterns: (a) Convolution: The center token aggregates information from a fixed local neighborhood. (b) Self-attention: The center token interacts with all tokens globally. (c) Swin: Local window-based self-attention. (d) VMamba: Global context is captured via directional sequential scanning (horizontal(black) and vertical(blue)).

selective scan operations to 2D image data by traversing patches along four directions and aggregating context across the spatial plane. It preserves global receptive fields and dynamic weighting while reducing complexity from quadratic to linear in the number of tokens.

Vim (Zhu et al., 2024a) proposes a pure state space model (SSM)-based vision backbone that replaces self-attention in VTs with bidirectional Mamba blocks to efficiently learn visual representations. Instead of using quadratic-cost attention, Vim treats an image as a sequence of patches with positional embeddings and applies bidirectional SSMs to capture global context and spatial relationships. Each Vim block normalizes token sequences, projects them for forward and backward SSM processing, and combines them with gating to compress visual features, yielding competitive accuracy while being significantly faster and more memory-efficient than standard VTs.

VideoMamba (Li et al., 2024a) is tailored for efficient video understanding that addresses both local redundancy and long-range temporal/spatial dependencies in video data. It extends bidirectional SSM scans into 3D spatiotemporal sequences, combines them with lightweight convolutions, and adds a novel self-distillation technique to improve scalability without large pre-training, where they first train a smaller, well-trained version of the model and then use it as a teacher to guide the training of the larger student model.

VSSD (Shi et al., 2025) extends SSM to better suit non-causal image data by reformulating how the state space duality (SSD) mechanism operates. It introduces a Non-Causal SSD formulation that removes dependencies on previous tokens, allowing every token to contribute to a global hidden state independently of ordering. It involves discarding the magnitude effects of interactions between hidden states and tokens while preserving relative weights, and integrating results from multi-scan strategies to achieve non-causal context integration with linear complexity.

SpatialMamba (Xiao et al., 2025a) advances prior Mamba-style vision SSMs by embedding structure-aware neighborhood connectivity directly into the latent state space rather than relying solely on scanning flattened 1D sequences. Spatial-Mamba first computes states using a unidirectional SSM scan, then applies a structure-aware state fusion module that uses dilated convolutions to fuse neighboring states in 2D, enriching each latent representation with local spatial context before producing outputs. This enables the model to better capture complex image spatial dependencies while maintaining linear complexity and strong performance.

On the other hand, 2DMamba (Zhang et al., 2025) preserves true 2D spatial structure throughout by scanning 2D grids directly. It models 2D image spatial context by extending the 1D Mamba selective SSM into a 2D selective scan framework that preserves image spatial continuity rather than flattening images into 1D sequences. It does this by applying a horizontal scan across rows followed by a vertical scan across columns for each state dimension, enabling efficient context aggregation over large 2D grids with high parallelism and linear complexity, making it efficient enough for tasks like giga-pixel whole slide image classification.

DefMamba (Liu et al., 2025) improves how SSMs process images by introducing a deformable scanning strategy and deformable state space blocks to dynamically adjust how image tokens are flattened and sequenced. Learns offsets that shift reference points and token indices toward semantically important regions, producing a structure-aware sequence that better preserves spatial information and object detail. It uses a multi-scale backbone with deformable Mamba blocks to adaptively prioritize informative features, resulting in stronger performance with linear complexity.

EfficientViM (Lee et al., 2025) redesigns the State Space Duality (SSD) layer to operate within compressed hidden states using a Hidden State Mixer SSD, rather than performing heavy channel mixing on full token sequences. By projecting image tokens into a smaller latent hidden state space, applying gating and mixing there, and then reconstructing outputs, it significantly reduces computational cost and memory-bound operations. It also includes multi-stage hidden state fusion, which aggregates representations from multiple stages to strengthen feature expressivity, achieving a superior speed–accuracy trade-off.

Mamba-adapter (Xie et al., 2025) addresses key weaknesses of vanilla SSMs when applied to images, like causal global context access, long-range forgetting of hidden states, and loss of spatial structure due to 2D-to-1D flattening. It adds adapter modules to the Mamba solving process. Adapter-T dynamically selects and aggregates important past hidden states to mitigate long-range forgetting, and Adapter-S applies multi-scale dilated convolution on the reshaped 2D outputs to introduce image inductive bias and enhance spatial locality. Both modules expand the effective context that the causal SSM can access without significantly increasing complexity.

5.2. Hybrid SSM

Hybrid models combine complementary architectural paradigms, most commonly convolutional neural networks, Transformers, and State Space Models, to leverage their respective strengths within a unified framework. While CNNs provide strong inductive biases for local spatial modeling and efficiency, Transformers excel at capturing long-range global dependencies through self-attention, and SSMs offer linear-complexity sequence modeling with efficient global context propagation. Hybrid designs strategically integrate these components to achieve improved accuracy–efficiency trade-offs. Such architectures have become increasingly prominent in recent vision research, as they balance scalability, computational efficiency, and representational power across diverse visual tasks.

Self-attention and SSM-based architectures differ fundamentally in how they model token interactions. Self-attention explicitly computes pairwise relationships, offering high expressiveness and flexibility at the cost of quadratic complexity. In contrast, SSMs rely on implicit global modeling through learned state transitions, achieving linear complexity but with more constrained interaction patterns.

MambaVision (Hatamizadeh and Kautz, 2025) introduces a hybrid vision backbone that combines the SSMs with the self-attention. It replaces the original causal convolution in Mamba with a standard convolution to remove directional constraints that are unnecessary for vision tasks. It also introduces a parallel symmetric branch without SSM, composed of an additional convolution and SiLU activation, to recover spatial information that may be limited by sequential SSM processing. The outputs of both branches are concatenated and linearly projected, enabling the model to jointly capture sequential and spatial features. Furthermore, the framework supports window-based self-attention to efficiently compute attention within local regions while maintaining scalability.

TranMamba (Zhang and Wan, 2025) is designed for single-image super-resolution, which interleaves Transformer layers with Mamba modules. It includes Transformer Aggregation Block and the Mamba Aggregation Block, each incorporating a Reparameterized Convolution (RepConv) to capture local spatial patterns alongside global cues. Additionally, a Reparameterized Spatial-Gate Feed-Forward Network (RepSGFN) augments the network's ability to mix spatial features adaptively within both hybrid blocks. During training, this structure enables the model to capture long-range interactions via self-attention and efficient global context via SSMs, while RepConv/RepSGFN modules improve local detail recovery crucial for HR image reconstruction.

MAP (Liu and Yi, 2025) proposes a novel pretraining strategy called Masked Autoregressive Pretraining (MAP). The core idea is that standard pretraining methods such as Masked Autoencoders (MAE) work well for pure Transformer modules but are not effective for Mamba modules, and vice versa for classic autoregressive pretraining. MAP combines the strengths of both by first randomly masking image patches and then reconstructing them autoregressively row-by-row, which enables the Transformer components to learn local bidirectional context through a local MAE objective and the Mamba SSM components to learn long contextual dependencies via autoregressive reconstruction aligned with their scanning order. This hierarchical pretraining framework significantly improves downstream performance on 2D and 3D vision tasks compared to single-strategy pretraining.

A2Mamba (Lou et al., 2025) integrates multi-scale self-attention with state space modeling to overcome limitations of simply stacking Transformers and SSMs. It proposes a Multi-scale Attention-augmented State Space Model (MASS) token mixer where multi-scale attention maps are used to spatially aggregate the hidden states of an SSM via a cross-attention-like mechanism, enhancing both 2D spatial dependencies and dynamic sequence modeling within a single mixer. This yields a pyramid backbone that jointly captures fine-grained local detail and long-range global context with sub-quadratic complexity across stages.

Mamba as a Bridge model (Zhang and Tan, 2025) proposes MFuser that combines Vision Foundation Models (VFM) and Vision–Language Models (VLM) for domain-generalized semantic segmentation without the high cost of full fine-tuning. MFuser consists of MVFuser and MTEnhancer. MVFuser is a Mamba-based co-adapter that takes concatenated token features from both VFM and VLM encoders and uses state-space modeling alongside parallel spatial processing to jointly refine sequential and spatial relationships. MTEnhancer is a hybrid Attention–Mamba module that refines text embeddings using visual priors to improve image–text alignment and semantic consistency. Both modules offer linear scalability with token length while combining fine-grained visual detail from VFMs and robust text grounding from VLMs for improved segmentation performance.

QMambaBSR (Di et al., 2025) introduces a burst image super-resolution framework that combines state space modeling with a query-guided attention mechanism tailored for aligning and aggregating information across multiple low-resolution frames. It uses Query State Space (QSS) blocks to propagate and integrate temporal/spatial information across burst images in a linear-complexity manner, while a lightweight cross-attention module computes affinities between a reference frame query and features from other burst frames to guide fusion. By combining SSM's efficient global context propagation with query-conditioned attention for precise frame correspondence, QMambaBSR achieves high-quality super-resolution with improved handling of motion and misalignment in burst sequences.

MaTVLM (Li et al., 2025) proposes a hybrid vision–language model that replaces a part of the standard Transformer decoder layers in a pretrained vision–language model (VLM) with Mamba-2 state space model (Dao and Gu) layers, striking a balance between efficiency and contextual understanding. It initializes Mamba-2 layers from corresponding Transformer attention weights to accelerate convergence, and then uses single-stage knowledge distillation from the original VLM to the hybrid model to retain performance while reducing computational burden.

Vamba (Ren et al., 2025) combines Mamba-2 state space model blocks with Transformer attention mechanisms to enable efficient understanding of long videos without reducing token counts. It replaces the quadratic causal self-attention over all video tokens in standard Transformer-based large multimodal models (LMMs) with Mamba-2 blocks that encode video tokens with linear complexity, while retaining attention mechanisms for text tokens via a mix of cross-attention and self-attention to update text features conditioned on video content. This reduces both memory and computation and allows Vamba to handle over 1024 frames on a single GPU with significant efficiency and competitive performance.

BMTNet (Zhou et al., 2025b) proposes a highly efficient binary neural network architecture designed for real-time demosaicing on edge devices. To capture both global context and local detail with minimal cost, BMTNet uses a hybrid Binarized Mamba-Transformer block that combines a binarized Mamba where non-critical projection layers are binarized but the core Selective Scan remains full precision, with a binarized Swin-style Transformer branch in parallel. This design efficiently models long-range dependencies via the state-space operator and local features via sliding window attention, significantly reducing parameters and computation while maintaining high demosaicing quality. An additional binarized global visual encoder supplies global priors to enhance context within the selective scan control matrices.

VL-Mamba (Qiao et al., 2024) is a multimodal large language model that uses a pretrained Mamba language model as the core LLM and introduces a MultiModal Connector (MMC) that bridges visual and textual modalities by applying a Vision Selective Scan (VSS) module to map image features into a sequential form compatible with the SSM backbone. A vision encoder produces visual tokens for initial feature extraction and leverages state-space scanning mechanisms to integrate multimodal context efficiently, avoiding full quadratic self-attention between all tokens.

MILA (Han et al., 2024) integrates selective SSM of Mamba and linear attention by two ways. First, it substitutes the entire Transformer block with Mamba's block architecture and second it replaces the attention sub-block with Mamba's block design while preserving the MLP block. It also shows that Mamba resembles a variant of linear attention Transformer with key differences such as an input gate, forget gate, residual shortcut, lack of attention normalization, single-head design, and modified block structure that together improve performance and efficiency.

This fusion model (Guo and Xie, 2025) integrates a Multi-Modal Fusion Attention (MMFA) module that uses frequency-domain and edge-guided priors to modulate standard QKV attention weights so the model focuses on semantically meaningful image regions, and a Bi-Directional Mamba Block (BDMB) that applies Mamba SSM scans along both horizontal and vertical axes to capture long-range spatial dependencies efficiently with linear complexity. Attention is tightly fused with the SSM components to balance fine-grained local restoration and global structural consistency in weather-degraded images.

Contrast (Urumbekov and Chen, 2025) architecture interleaves Mamba SSM blocks with Transformer attention blocks, overlapping cross-attention blocks that refine cross-window and long-range spatial

Table 2
Representative Mamba-based vision models. N = spatial tokens, T = temporal length.

Model	Category	Scan/Mixing	Complexity
ViM (Vision Mamba)	SSM-based	1D bi-scan	$\mathcal{O}(N)$
VMamba	SSM-based	4-dir scan (CSM)	$\mathcal{O}(N)$
VideoMamba	SSM-based	Spatiotemporal scan	$\mathcal{O}(NT)$
VSSD	SSM-based	Non-Causal SSD	$\mathcal{O}(N)$
SpatialMamba	SSM-based	Spatial fusion scan	$\mathcal{O}(N)$
2DMamba	SSM-based	2D row+col scan	$\mathcal{O}(N)$
DefMamba	SSM-based	Deformable scan	$\mathcal{O}(N)$
EfficientViM	SSM-based	Compressed-state scan	$\mathcal{O}(N)$
MambaVision	Hybrid (SSM+Attn)	Parallel SSM+Attn	Hybrid
A2Mamba	Hybrid (SSM+Attn)	Multi-scale Attn	Hybrid
VL-Mamba	Hybrid (SSM+MM)	VSS backbone	Hybrid

dependencies to capture fine-grained pixel relationships critical for high-quality image reconstruction. It replaces standard MLPs with Spatial Gated Feed-Forward Networks (SGFN) for enhanced local processing.

Wang et al. (2025a) uses frozen pre-trained models to extract spatial and grid-based features, alongside cross-domain retrieval for semantically aligned text descriptions. The hybrid encoder employs dynamic gating to combine the strengths of attention and state space models. During decoding, the feature aggregation module adaptively fuses multimodal features via learnable weights. Table 2 summarizes representative Mamba-based vision architectures and their scanning strategies and computational complexity.

Advantages and limitations

State space model (SSM)–based architectures, including Mamba and its variants, offer several compelling advantages over traditional self-attention-based vision models. By replacing the quadratic complexity of attention with linear-time structured state space operations, SSMs enable efficient processing of long sequences, substantially reducing computational and memory costs as sequence length increases. This property is particularly attractive for high-resolution inputs and dense prediction tasks, where token counts are large. Moreover, SSMs incorporate data-dependent parameterization, enabling content-aware reasoning through selective retention or forgetting of information, an effect conceptually analogous to the query–key mechanism in attention and beneficial for salient region modeling in vision.

However, these efficiencies come with inherent trade-offs. As highlighted by MambaOut (Yu and Wang, 2025), the recurrent SSM core in Mamba relies on a fixed-size hidden state, which compresses past information and is therefore lossy compared to the explicit key–value memory in attention mechanisms. While this recurrent design is well suited to long-sequence and autoregressive settings, many standard vision benchmarks, such as ImageNet classification, exhibit neither extreme sequence length nor causal structure. In such cases, the advantages of SSMs diminish as MambaOut demonstrates that a simpler Gated CNN backbone can outperform visual Mamba models incorporating SSMs on ImageNet. Nevertheless, this advantage does not consistently extend to detection and segmentation tasks, which better reflect long-sequence characteristics, suggesting that SSMs remain beneficial in those contexts.

Architecturally, pure SSM-based vision models (e.g., Vim (Zhu et al., 2024a) and VMamba (Liu et al., 2024b)) introduce additional complexity when adapting 1D state space formulations to 2D image data. As images are processed through sequential scanning schemes, these models require bidirectional or multi-directional scanning strategies to mitigate directional bias and approximate spatial interactions. In contrast, CNNs natively encode 2D locality through convolution, and VTs, while operating on patch sequences, rely on global attention rather than recurrent scanning and therefore avoid such directional adaptations.

Hybrid Mamba–Transformer architectures aim to combine the linear efficiency of SSMs for long-range compression with the expressive flexibility of attention, with specialized pretraining strategies (e.g., MAP (Liu and Yi, 2025)) reporting gains. However, these hybrids increase architectural complexity and may introduce training instability or delicate trade-offs between efficiency and expressivity. Furthermore, when applied directly to 2D vision tasks, SSMs can exhibit weaker spatial structure modeling and long-range forgetting, often requiring additional mechanisms such as spatial fusion modules or convolutional inductive biases, to achieve competitive performance.

Taken together, SSM-based and hybrid architectures represent prom ising alternatives or complementary techniques to self-attention, but they do not constitute a universal replacement. Their effectiveness depends critically on task characteristics, sequence length, spatial structure requirements, and computational constraints.

6. Multimodal and generative VTs

Vision Transformers serve as critical components within larger systems such as visual encoders for multimodal LLMs, text-to-image generation frameworks, and unified models that jointly optimize for understanding and generation. This shift fundamentally changes how we evaluate backbone architectures. A ViT optimized purely for ImageNet accuracy may exhibit poor token efficiency or weak semantic alignment. In this section, we broaden our discussion to examine how the self-attention innovations impact performance when VTs are deployed as components in modern multimodal and generative systems.

6.1. ViT for multimodal LLMs

Multimodal LLMs use architecture of a pretrained vision encoder such as ViT-Huge to process images into visual tokens, a projector aligns these tokens with the text embedding space, and an LLM performs autoregressive generation conditioned on both modalities (Zamini and Shukla, 2025). While efficiency-focused variants discussed in Section 3, use different techniques to save compute, these techniques often destroy the dense spatial relationships needed for linguistic reasoning. ViT-Huge is preferred in MLLMs because its global self-attention preserves the semantics and tokens that encapsulate complex scene concepts, allowing the LLM to reason about objects, text (OCR), and spatial logic more effectively.

CLIP (Radford et al., 2021) variants dominated this role due to their contrastive pretraining, which aligns images and text in a shared semantic space (Xiao et al., 2025b). However, CLIP's softmax-based contrastive loss requires global normalization across the batch, limiting scalability and batch size efficiency. SigLIP (Zhai et al., 2023) replaces softmax with a sigmoid-based binary classification objective applied independently to each image–text pair. It solves this by using a pairwise sigmoid loss, which decouples the batch size from the objective. This allows the self-attention mechanism to be trained on large number of image–text pairs, leading to the zero-shot transfer capabilities seen in SigLIP-2 (Tschannen et al., 2025) and Gemma 3 (Team et al., 2025), where the backbone generalizes to almost any visual domain without fine-tuning.

LLaVA (Liu et al., 2023a) use CLIP to extract visual features, project them into the language model's embedding space, and then fine-tune an LLM with automatically generated multimodal instruction-following data so that it can answer diverse vision–language queries and follow human instructions. The architecture uses cross-modal attention between projected visual tokens and the language model's auto-regressive token stream during decoding. It is later strengthened in Liu et al. (2024a) by using a higher-resolution CLIP backbone with an MLP projection instead of a linear projection and adding academic VQA datasets with response-formatting prompts which leads to improved performance. Further, LLaVA-MORE (Cocchi et al., 2025) extend the standard LLaVA architecture by combining the visual encoder with

various LLMs, ranging from small- to medium-scale models. It keeps ViT self-attention for image encoding and uses cross-modal attention in the LLM adapter to fuse vision and language. Unlike the original LLaVA, it focuses on a systematic, controlled study of backbone and LLM choices rather than new architectural changes, offering practical guidance for effective multimodal model design.

BLIP (Li et al., 2022a) adopts an end-to-end Transformer architecture with a ViT encoder and a multimodal encoder–decoder trained using multiple objectives such as image–text contrastive learning, where self-attention and cross-attention mechanisms are jointly optimized. BLIP-2 (Li et al., 2023b) improves efficiency by freezing a pretrained ViT and LLM and introducing a lightweight Q-Former, which uses learnable queries and cross-attention to extract relevant visual features for alignment, significantly reducing training cost. In contrast, BLIP-3 (Xue et al., 2025) further simplifies and scales the framework by removing the Q-Former and employing a vision token sampler to compress image tokens before feeding them directly into the LLM, training the entire system with a unified autoregressive objective, enabling more scalable large-scale multimodal modeling.

ViT-Lens (Lei et al., 2024) introduces a modality-specific Lens that maps raw inputs from any modality into an intermediate embedding space compatible with a frozen pretrained ViT, enabling the ViT to act as a shared omni-modal processor. The modality encoder consists of a lightweight embedding module and a Lens network that projects modality tokens into a latent space before feeding them into selected frozen ViT layers whose self-attention then computes contextual interactions among tokens. Representations are aligned to a predefined feature space using contrastive learning, enabling improved generalization even with limited data. Unlike ViT models trained on images only, ViT-Lens enables any modality to be processed through shared ViT attention layers. It supports emergent any-modality captioning, QA, and generation in a zero-shot manner when plugged into multimodal foundation models.

VCoder (Jain et al., 2024) is designed to improve low-level visual perception in MLLMs, which existing systems like LLaVA struggle with despite strong high-level reasoning. Its vision encoder takes the RGB image and perception-oriented inputs, such as segmentation maps and depth maps, and projects them into the LLM's embedding space via lightweight encoder branches that feed into the language model to improve visual grounding and object perception.

Qwen3-VL (Bai et al., 2025) is designed to handle extremely long interleaved contexts. It maintains a three-module design comprising a vision encoder, an MLP-based merger to compress and align visual features, and an LLM backbone that processes the combined multimodal token stream. It includes an interleaved MRoPE positional encoding for balanced spatial–temporal modeling, a DeepStack mechanism that fuses multi-level visual features into corresponding LLM layers for tighter alignment, and text-based timestamp tokens for precise video grounding.

VisionLLM (Wu et al., 2024) presents an information transmission mechanism that acts as a flexible intermediary between the shared multimodal backbone and multiple task-specific decoders, allowing effective gradient flow and task signaling during end-to-end multi-task training. It supports task-conditional decoding via shared parameters and prompt signals. It integrates cross-modal attention to fuse visual and language features and passes context through the super links to task heads, enabling both intra-modal and inter-modal interaction in a scalable way.

Advantages and limitations

The Transformer architecture is modality-agnostic, enabling unified processing of vision and language. VTs patchify images in a way that naturally aligns with LLM tokenization, supporting early fusion where image patches and text tokens interact from the first layer. This yields richer cross-modal interactions and, as scaling studies show, more compute-efficient performance than stitched late-fusion models, with advantages growing at scale (Shukor et al., 2025). The shift from CLIP (Radford et al., 2021) to SigLIP (Zhai et al., 2023) improved contrastive pretraining by replacing softmax with a sigmoid loss, removing global batch normalization and enabling flexible batch sizes and better scalability. Pretrained VTs learn structured, semantically clustered embedding spaces, making them strong frozen encoders in multimodal systems like LLaVA (Liu et al., 2023a). Compared to classification-trained VTs, SigLIP-style encoders provide better zero- and few-shot transfer, supporting a unified sequence modeling framework and simpler cross-attention integration across modalities.

Despite improved token efficiency, MLLMs remain computationally constrained as their quadratic complexity in sequence length makes even moderate visual token counts costly. For example, 256 tokens from a 224×224 image already has a reasonable inference cost, while 4K images can exceed 100,000 tokens, making them expensive, even with optimized attention (Zamini and Shukla, 2025). Scaling law analyses (Shukor et al., 2025) show that separate vision and language backbones duplicate resources instead of optimizing jointly, while late fusion limits cross-modal interaction to shallow adapter layers, missing richer early-fusion dynamics. As a result, for the same compute budget, native multimodal models consistently outperform stitched architectures like LLaVA (Liu et al., 2023a), indicating that many current MLLMs operate below their optimal efficiency frontier.

6.2. ViTs for generative modeling

Beyond multimodal understanding, self-attention backbones are increasingly deployed in generative vision systems.

6.2.1. Diffusion VTs

U-ViT (Bao et al., 2023) is a ViT-based architecture tailored for score-based diffusion models that replaces the CNN U-Net with a Transformer treating noisy image patches, time embeddings, and text embeddings as a unified token sequence. It augments the standard ViT with U-Net-style long skip connections to preserve low-level features crucial for pixel-level noise prediction, mitigating the loss of local detail common in vanilla ViTs for image synthesis. Time and conditioning signals are concatenated directly with token embeddings, enabling joint attention over spatial and auxiliary information. An optional lightweight convolutional head further improves visual quality by reducing Transformer-induced artifacts.

DiT (Peebles and Xie, 2023) also replaces U-Net with Transformer encoder architectures operating on latent image patches within a latent diffusion framework. It takes a VAE latent representation and patchifies it into a sequence of tokens, adds positional embeddings, and processes these via stacks of MHSA. Conditional information like timestep and class labels is integrated through in-context tokens, cross-attention, and adaptive LayerNorm. FiT (Lu et al., 2024) addresses the fixed-resolution limitation of models like DiT by representing images as variable-length token sequences, enabling arbitrary resolutions. Instead of resizing or cropping to a fixed grid, it encodes image latents, patchifies them into dynamic token sequences, and pads to a maximum length for batching. During inference, tokens are reshaped to the target resolution. FiT employs Masked MHSA to prevent padding tokens from affecting attention, adopts 2D Rotary Positional Embeddings (2D-RoPE) for resolution-insensitive spatial awareness, and replaces standard MLPs with SwiGLU activations for improved efficiency.

SiT (Ma et al., 2024) formulates image generation as a continuous dynamical transport process that flexibly interpolates between data and noise, instead of following a fixed forward diffusion schedule. The interpolant is treated as a configurable design choice, allowing independent control over time parameterization, mixing curves, training objectives, and sampling methods. By decoupling and tuning these components, SiT simplifies learning and consistently improves generative performance over the DiT baseline without modifying the Transformer backbone.

MDT (Gao et al., 2023) enhances contextual relation learning among semantic image parts by masking latent tokens during training and predicting them from unmasked ones within the diffusion denoising process. It adds noise to a VAE latent and patchifying this into tokens. A random subset of these tokens are masked such that the model learns to reconstruct missing parts from context, explicitly forcing richer relational modeling among object parts than typical diffusion training does.

SD-DiT (Zhu et al., 2024c) accelerates convergence and improves generative quality by integrating self-supervised discrimination into training. It adopts a teacher–student framework (in the same vein as DeiT (Touvron et al., 2021a)) that decouples DiT encoder and decoder objectives instead of jointly optimizing mask reconstruction. The encoder performs discriminative learning by aligning embeddings of positive pairs from different noisy versions of the same image along the diffusion Probability Flow ODE, using a novel joint-embedding loss, while the decoder focuses solely on the diffusion objective. The teacher encoder, updated via an exponential moving average of the student, provides stable discrimination targets.

GenTron (Chen et al., 2024) also extends DiT by integrating text conditioning into the Transformer through concatenating text embeddings to token sequences, cross-attention conditioning, and joint projection spaces, enabling fine-grained control of generation. It also scales the Transformer parameters massively to improve the visual quality. It further adapts attention for temporal modeling in video by either augmenting tokens with frame position embeddings or by designing attention blocks that attend across both spatial and temporal token dimensions.

DiffiT (Hatamizadeh et al., 2024b) introduces a Time-Dependent MHSA tailored to joint spatial–temporal modeling in the iterative denoising process. It modulates the query, key, and value projections as a function of the noise level, enabling the attention layers to adapt their behavior dynamically over the sampling trajectory and better capture how spatial relationships should change as noise is removed. It includes learned temporal components for Q, K, and V, effectively conditioning self-attention on time without requiring separate adaptive normalization layers for conditioning.

6.2.2. GANs with VTs

TransGAN (Jiang et al., 2021a) replaces all convolutions in both the generator and discriminator with Transformers, showing that pure attention-based GANs can match or surpass CNN-based models on standard benchmarks. Generator and discriminator networks represent images as sequences of patch tokens processed via MHSA. To reduce computational cost while preserving long-range dependencies, it introduces grid self-attention, which applies attention within structured spatial blocks. TranGAN also employs a multi-scale discriminator that hierarchically attends to different resolutions and adopts tailored training strategies to stabilize adversarial learning and alleviate Transformers' data inefficiency.

ViTGAN (Lee et al., 2022) adopts a pure ViT architecture and enhances stability with several targeted modifications. It uses L2 attention, based on Euclidean-distance similarity with tied query and key projections, along with improved spectral normalization to enforce Lipschitz continuity in the Transformer discriminator. Overlapping image patches are introduced to reduce grid sensitivity and better capture local structure, while convolutional projections in the Q/K/V layers inject spatial priors into attention. In the generator, self-modulated layer normalization conditions Transformer blocks on the input latent vector, improving convergence and training stability.

StyleSwin (Zhang et al., 2022a) adopts Swin blocks into a style-based GAN architecture. It introduces a double-attention mechanism in each Transformer block that attends to both standard and shifted local windows, expanding contextual modeling without much computational overhead. To compensate for the loss of absolute position in windowed attention, sinusoidal positional encodings are added at every layer to preserve global coherence, essential for high-resolution synthesis. The generator adopts a style-based modulation scheme, injecting latent codes across layers to control multi-scale features, and employs a wavelet-domain discriminator to enhance spectral consistency in the outputs.

6.2.3. Autoregressive VTs

VAR (Tian et al., 2024) rethinks visual generation as a next-scale prediction instead of a traditional raster-scan next-token prediction. Images are first encoded with a multi-scale VQ-VAE tokenizer into a sequence of increasingly higher-resolution token maps. VAR then trains the model where each entire token map is predicted from all preceding scales. It uses a causal mask designed for hierarchical autoregressive generation rather than the global bidirectional attention typically used in ViT classification. It performs coarse-to-fine autoregression over spatial scales while FAR (Yu et al., 2025) generates the global low-frequency structure, then progressively add high-frequency details using a continuous tokenizer.

Li et al. (2024c) eliminates discrete tokenizers by operating directly in a continuous representation space, combining autoregressive modeling with diffusion-based per-token probability estimation to generate high-fidelity images without vector quantization. It introduces a Diffusion Loss, where a lightweight denoising network models each token's continuous conditional distribution via a diffusion process, enabling flexible sampling without discrete categories. The framework also supports generalized masked autoregressive variants, allowing subsets of tokens to be predicted jointly while maintaining causal attention constraints.

Wang et al. (2025b) decouples discretization from tokenizer learning by first training a continuous VAE tokenizer to produce high-fidelity continuous latents, then applying a post-training, dimension-wise quantization scheme to convert these continuous features into discrete tokens without the non-differentiable training pitfalls of traditional quantization-during-training schemes. This results in a very large discrete token vocabulary, which is addressed with a lightweight autoregressive prediction mechanism that factorizes the prediction of each high-dimensional token into a sequence of dimension-wise predictions to efficiently model inter-dimensional dependencies.

GigaTok (Xiong et al., 2025) shows that increasing tokenizer size improves reconstruction but degrades autoregressive generation quality because the latent space becomes too complex for the generator to learn effectively. To address this, it introduces semantic regularization that aligns the tokenizer's latent features with semantically consistent features from a pretrained visual encoder, constraining latent complexity and stabilizing scaling. The tokenizer is a vector-quantized encoder–decoder with a hybrid CNN–Transformer backbone and explores three scaling practices. 1D tokenizers with flattened spatial sequences scale better than 2D grid tokenizers, prioritizing decoder scaling improves downstream generation compared to symmetric scaling, and intensive entropy loss stabilizes large-scale training at 3B parameters by encouraging uniform codebook usage.

LlamaGen (Sun et al., 2024b), based on LLaMA architecture, uses an image tokenizer that compresses an image into a discrete sequence of tokens via a codebook, enabling a raster-order sequence suitable for LLM-style prediction. The architecture is scaled across parameter counts of 111M to 3.1B and show that larger models can yield better fidelity.

6.2.4. Hybrid models

MADFormer (Chen et al., 2026) proposes a hybrid Transformer that combines autoregressive and diffusion modeling for continuous image generation. It partitions images into spatial blocks, applies autoregressive conditioning across blocks to capture global structure, and performs diffusion-based denoising within each block to refine continuous latents, balancing long-range dependency with fine-grained detail.

HART (Tang et al., 2025) addresses the reconstruction limitations of VAR tokenizers by decomposing continuous VAE latents into discrete tokens and continuous residual tokens, where discrete tokens represents the coarse global structure, modeled by a scalable autoregressive Transformer while continuous ones capture fine-detail information, learned via a lightweight residual diffusion module, improving reconstruction FID and generation FID.

AutoRegressive Distillation (Kim et al., 2025) accelerates Diffusion Transformers by replacing iterative sampling with a history-aware autoregressive predictor conditioned on the full ODE trajectory, reducing exposure bias. The student model learns to predict the next state from the current noisy input and its trajectory history using a modified DiT backbone with token-wise time embeddings and causal attention across trajectory states. Architectural variants balance efficiency and information flow, enabling high-quality generation with significantly fewer steps.

ViT-VQGAN (Yu et al., 2022) replaces convolutional quantizers with ViT-based VQGAN and then trains an autoregressive Transformer on the discrete image tokens for generation and representation learning. First, the encoder transforms an input image into a grid of discrete latent tokens then is trained autoregressively to predict the next token in the rasterized latent sequence.

Transfusion (Zhou et al., 2025a) trains a single Transformer to handle both discrete text and continuous images by combining next-token prediction with diffusion-based generation within the same architecture. It processes images as sequences of continuous patch vectors and interleaves them with text tokens in a mixed modality sequence, supervised simultaneously with a causal next-token loss for text and a diffusion loss for images.

Advantages and limitations

Diffusion frameworks such as SiT (Ma et al., 2024) introduce gating mechanisms that regulate information flow through residual branches, improved normalization strategies such as LayerScale and adaptive normalization, and learnable tokens that adapt to generative requirements beyond fixed positional encodings. Compared to CNN generators, ViT-based generators capture long-range dependencies more directly, improving global coherence and consistent relationships between distant objects. In adversarial settings, adding global receptive fields to the discriminator can give more complete consistency feedback and improve dynamics and sample quality. In autoregressive models, the shared Transformer structure allows language modeling techniques to be directly extended to visual token prediction. Self-attention and cross-attention enable conditioning on text, layouts, or segmentation maps, core to text-to-image diffusion and Transformer diffusion architectures (Chen et al., 2024; Peebles and Xie, 2023), supporting controllable, compositional generation. Autoregressive ViTs over joint image–text token streams can, in principle, do in-context conditioning from example pairs in the prompt without gradient updates, though few-shot vision generalization is typically weaker and more scale-sensitive than in LLMs. When combined with discrete tokenizers such as VQ-VAE or VQ-GAN, they operate over quantized visual tokens and can use standard LM objectives and sampling.

At inference, diffusion remains expensive because it requires many denoising steps, each involving a full forward pass. Even with linear-attention variants, dozens of steps multiply cost versus single-pass models. Accelerated samplers like DPM-Solver (Lu et al., 2022a) reduce the number of function evaluations but introduce trade-offs, including potential fidelity degradation at very low step counts, sensitivity to noise schedules, solver instability, and additional hyperparameter tuning to balance speed and quality. Unified encoders also face a representation tension as CLIP-style contrastive training encourages invariance to fine detail, while high-fidelity synthesis needs precise spatial and texture reconstruction, partly mitigated by scale and architecture, but the trade-off remains a practical consideration. These models are generally less prone to severe GAN-style mode collapse due to likelihood-based training, yet diversity can still degrade or overfit under limited and imbalanced data. ViT-GANs further demand heavy GAN- and Transformer-specific hyperparameter tuning and see less production uptake. Autoregressive models are costly at high resolution because tokens are generated sequentially, even with KV caching, runtime grows linearly with token count, errors compound as early mistakes propagate with limited global correction, and raster-scan ordering imposes an arbitrary directional bias, preventing conditioning on future spatial context, which can hurt global coherence, whereas diffusion refines all spatial tokens in parallel each step.

Table 3
Representative attention and token-level interpretability methods.

Method	Approach	Key mechanism
LeGrad	Gradient	Gradients w.r.t. attention maps
AttnCAM	Gradient	Class-gradient weighted attention
AttnLRP	Relevance Prop.	Deep Taylor decomposition of QK–V
TIS	Perturbation	Random token subset sampling
MTE	Perturbation	Multiplex token sampling
ICE/ICEv2	Learning-based	Post-hoc patch classifier
CLIP	Linguistic	Additive head-wise embedding split

7. Interpretability and explainability in vision transformers

Although Vision Transformer research has largely emphasized architectural efficiency and performance, interpretability has become equally critical. The global receptive field and token mixing of self-attention make VT decisions difficult to understand. Early approaches, such as attention visualization and rollout (Abnar and Zuidema, 2020), traced information flow by visualizing attention maps under the assumption that attention weights indicate token importance. However, Jain and Wallace (2019) showed that attention weights are not necessarily faithful explanations, as different attention distributions can yield similar outputs. This insight has driven the development of more rigorous, causally grounded explainability methods tailored to Transformer-based models. For example, LeGrad (Bousselham et al., 2025) computes the gradient of the decision score with respect to the self-attention maps at each Transformer layer which quantifies how sensitive the final output is to perturbations in individual attention weights and thus serves as the attribution signal. These layerwise gradients are then aggregated across all layers and combined with intermediate and final token activations to produce a unified relevance heatmap that more accurately reflects decision-critical regions while suppressing artifacts such as high attention tokens that carry little semantic relevance. By clipping negative gradients and summing the layer signals without additional normalization, LeGrad provides an explainability technique that works across ViT backbones. AttentionCAM (Leem and Seo, 2024) is another gradient-based explainability method that combines back-propagated class gradients with self-attention information to generate high-semantic visual explanations of a ViT's decisions. It computes the gradient of the classification score with respect to intermediate self-attention modules, capturing how changes in each self-attention layer influence final predictions, and then weights those gradients by normalized self-attention scores. It yields explainable heatmaps that reflect both the model's learned patch interactions and their influence on classification outputs.

GMAR (Jo et al., 2025) improves classic attention rollout by weighting attention heads with gradient-based importance. It computes the gradient of the target class logit with respect to each head's output to measure its influence, then uses these weights during recursive attention rollout across layers. The resulting aggregated map better captures hierarchical attention and class-specific sensitivity, providing more faithful interpretability than standard rollout.

Instead of perturbing pixels or relying on raw attention weights, Transformer Input Sampling (TIS) (Englebert et al., 2023) generates

binary masks over token embeddings after linear projection and positional encoding, but before self-attention, removing subsets of tokens entirely to avoid outlier distortions. Masks are formed by clustering token activations across layers, sampling representative patch subsets, and scoring each subset via forward passes. Class relevance weights derived from output logits are used to compute a normalized weighted sum of masks, producing the final saliency map. By perturbing tokens rather than pixels, TIS better aligns with ViT sequence processing and yields spatial relevance consistent with model predictions.

Gandelsman et al. (2024) analyzes how the CLIP image encoder forms its final embedding by decomposing it into additive contributions from patches, layers, and attention heads, leveraging the residual architecture and joint vision–language space for semantic interpretability. For each head, a TextSpan algorithm is applied that greedily selects text embeddings spanning the head's output subspace, assigning natural language labels to its principal directions. Token contributions are also spatially decomposed to produce heatmaps aligned with specific text directions. This text-grounded analysis reveals that many heads capture interpretable properties and implicitly localize meaningful features. These insights are further used to remove spurious features and build a zero-shot segmenter, showing that attention-based interpretability can both diagnose and improve model behavior.

AttnLRP (Achtibat et al., 2024) adapts Layer-Wise Relevance Propagation to attention layers for fine-grained attribution. Using a Deep Taylor Decomposition framework, it backpropagates the target logit's relevance through all Transformer operations, including query–key products, softmax, and layer normalization, while conserving relevance at each layer. By modeling self-attention as a decomposable graph, it distributes relevance across heads and layers in a single backward pass, revealing each input patch's and latent neuron's contribution to the prediction, achieving higher faithfulness and efficiency than gradient- or attention-only methods.

ICE (Choi et al., 2023) reframes ViT explainability by directly learning class labels for each patch token rather than deriving saliency from class token or gradient. It uses the output patch embeddings for each patch and trains a lightweight classifier over these embeddings with an adversarial normalization loss. By treating all patches initially as background and gradually learning patch-wise class assignments guided by the model's outputs, ICE produces high-resolution, class-specific activation maps that distinguish foreground object regions from background and thus yield explainability visualizations. ICEv2 (Choi et al., 2025) reduces fine-tuning to only the last few encoder layers, redesigns the MLP head (depth/width) for better generalization, and optimizes training hyperparameters across model sizes to preserve the original ViT weights, producing more faithful, comprehensive explainability maps. Multiplex Transformer Explainer (MTE) (Marchetti et al., 2025) is a post-hoc explainability framework that models self-attention across all Transformer layers as a multiplex network. Each layer's attention matrix forms a graph layer where nodes represent image patches and edges encode attention-based interactions. By applying centrality measures and cross-layer aggregation, MTE captures nonlinear patch relationships throughout the model. It then generates structure-informed mask perturbations, evaluates them via forward passes, and weights the masks by class confidence to produce heatmaps highlighting patches most influential to the prediction.

Such methods are typically evaluated using insertion, deletion, mean, and pointing game scores, and consistently outperform earlier attention-based explanations. For comprehensive overviews, see recent surveys on VT explainability (Fantozzi and Naldi, 2024; Kazmierczak et al., 2025). Integrating these techniques with efficient self-attention variants is a promising step toward VTs that are both high-performing and trustworthy. Table 3 compares representative interpretability methods based on their underlying approach and key mechanism.

8. Performance and efficiency

This section analyzes emerging efficiency and accuracy trade-offs across architectural families. By grouping models by parameter scale and comparing their positions on the accuracy and FLOPs frontier, we identify recurring structural patterns that define high-performing Vision Transformer designs. A comparative analysis of the best-performing models in terms of accuracy and model size is presented in Table 6. The models are categorized into five groups based on their parameter sizes, extra-tiny (≤10M), tiny (11M–22M), small (23M–50M), base (51M–88M) and large (>88M), (M represents million). The survey reports Top-1 accuracy, parameter count, and FLOPs, using results directly from the original papers. When available, ImageNet-1K results under commonly used training settings are chosen to ensure fair comparison. Model size designations (e.g., Tiny, Small, Base, Large) vary across papers and follow each author's convention. For analytical clarity, this survey occasionally groups or references models using a unified size categorization, which may differ from the original designations.

From each group, the top 10 models are selected based on Top-1 accuracy. The leading performers in each group are CoaT (Xu et al., 2021a) (extra-tiny), Pale (Wu et al., 2022a) (tiny), DualViT (Yao et al., 2023) (small), LV-ViT (Jiang et al., 2021b) (base) and VOLO (Yuan et al., 2022) (large). CoaT (Xu et al., 2021a) and Pale (Wu et al., 2022a) benefit from multi-scale and axis-aware attention, which improves token interaction without incurring full quadratic cost. DualViT (Yao et al., 2023) and GCViT (Hatamizadeh et al., 2023) leverage explicit local–global interaction mechanisms, allowing semantic priors to guide fine-grained feature refinement. DualViT (Yao et al., 2023) leverages explicit local–global interaction mechanisms, allowing semantic priors to guide fine-grained feature refinement. LV-ViT LV-ViT (Jiang et al., 2021b) improves representation quality at the tokenization stage through progressive aggregation or token labeling, enhancing data efficiency in mid-scale regimes. VOLO (Yuan et al., 2022) incorporates outlook attention that preserves spatial detail while controlling computation at higher resolutions.

However, when evaluating accuracy per parameter, the most performant models are LVT (Yang et al., 2022a), T2T (Yuan et al., 2021c), Wave-ViT (Yao et al., 2022), and GCViT (Hatamizadeh et al., 2023) for their respective size groups. GCViT (Hatamizadeh et al., 2023) performs due to integrating local and global information which enables high-level semantic knowledge to direct the refinement of detailed features. T2T (Yuan et al., 2021c) enhances feature representation during tokenization by gradually aggregating information. LVT (Yang et al., 2022a) enhances low-level feature representations in the initial stage by integrating self-attention mechanisms into the convolutional kernel. Wave-ViT (Yao et al., 2022) preserves details using wavelet-based decomposition.

Models such as EfficientViT (Liu et al., 2023b), CMT (Guo et al., 2022a), and EdgeViT (Pan et al., 2022a) achieve strong FLOPs-normalized performance primarily due to structured locality constraints and hardware-aware design choices that reduce redundant global interactions. In terms of overall cost-effectiveness, EdgeViT (Pan et al., 2022a), EfficinetViT (Liu et al., 2023b), EfficinetFormerV2 (Li et al., 2023e), FasterViT (Hatamizadeh et al., 2024a) and NAT (Hassani et al., 2023) are the top models in each category. FasterViT (Hatamizadeh et al., 2024a) and NAT (Hassani et al., 2023) show that limiting attention to spatial neighborhoods while allowing controlled cross-region communication leads to strong cost and accuracy trade-offs. Overall, frontier models combine hierarchical structure, localized attention, and selective global aggregation instead of relying only on attention reformulation or brute force scaling. Table 5 also shows the attention categories of top-performing models.

Table 4 highlights the top models evaluated across different input resolutions. DaViT (Ding et al., 2022) leads with 90.4% accuracy,

Table 4
Comparative analysis between different vision Transformer models grouped by image size, highlighting their parameter count, operations (GFLOPs), and Top-1% accuracy on ImageNet-1K. Top models with the highest Top-1% accuracy from each group are presented.

Image size	Method	Acc. (%)	Params (M)	FLOPs (G)
512 × 512	DaViT (Ding et al., 2022)	90.4	1437	1038
	VOLO (Yuan et al., 2022)	87.1	296	412
	MaxViT (Tu et al., 2022)	86.7	212	245
	LV-ViT (Jiang et al., 2021b)	86.4	151	214.8
448 × 448	KVT (Wang et al., 2022a)	86.5	86	67.9
	Refiner (Zhou et al., 2021b)	85.9	81	98
384 × 384	DualViT (Yao et al., 2023)	86.5	73	60.5
	MaxViT (Tu et al., 2022)	86.4	212	34.3
	UniFormer (Li et al., 2023c)	86.3	100	39.2
	DeepViT (Zhou et al., 2021a)	86.1	270.9	173.3
	XCiT (Ali et al., 2021)	86.0	189	417.8

Table 5
Comparative analysis between different vision Transformer models grouped by attention mechanism categories, highlighting their parameter count, operations (GFLOPs), and Top-1% accuracy on ImageNet-1K. The top three Base models with the highest Top-1% accuracy from each group are demonstrated.

Category	Method	Acc. (%)	Params (M)	FLOPs (G)
Axial	CSWin (Dong et al., 2022)	84.2	78	15
	DiagSWin (Li et al., 2024b)	84.4	46	9.0
	Pale (Wu et al., 2022a)	85.0	48	9
Convolution	UniFormer (Li et al., 2023c)	85.1	50	8.3
	Inception (Si et al., 2022)	84.6	48	9.4
	CMT (Guo et al., 2022a)	84.5	45.7	9.3
Cross	DualViT (Yao et al., 2023)	85.2	42.6	9.3
	CrossViT (Chen et al., 2021)	82.8	44.3	9.5
	CAT (Lin et al., 2022)	81.8	37	5.9
Formula	Wave-ViT (Yao et al., 2022)	84.8	34	7.2
	Castling-ViT (You et al., 2023)	84.6	34.7	7.0
	MiniViT (Zhang et al., 2022c)	84.3	46	31.4
Hierarchical	P2T (Wu et al., 2022b)	83.5	36.1	6.5
	MViT (Fan et al., 2021)	84.0	72.7	14.7
	FasterViT (Hatamizadeh et al., 2024a)	83.2	53.4	5.3
Mixed	DaViT (Ding et al., 2022)	84.2	49.7	8.8
	LST (Zhu et al., 2021)	83.8	39.8	8.7
	MaxViT (Tu et al., 2022)	83.6	31	5.6
Tokens	BiFormer (Zhu et al., 2023)	85.4	58	9.8
	DGT (Liu et al., 2022)	84.6	52	9.4
	Shunted (Ren et al., 2022)	84.0	39.6	8.1
Windows	VSA (Zhang et al., 2022d)	83.8	50	8.9
	Shuffle (Huang et al., 2021)	83.5	50	8.9
	HRFormer (Yuan et al., 2021d)	82.8	50.3	13.7

though it comes with high parameter and computational costs. Table 5 demonstrates the top-performing models in each attention category, highlighting that models using convolution-based attention or reformulated attention approaches generally outperform others.

Furthermore, Fig. 7 demonstrates the ImageNet top-1% accuracy with FLOPs and parameters count and four different graphs represent four sizes of models based on parameters count. Overall, the graphs and tables reveal that larger models achieve higher accuracy, while smaller ones trade performance for efficiency. It is important to note that FLOPs and model size are not reliable indicators of latency or memory consumption (Bello et al., 2021).

9. Challenges & future direction

ViT models have demonstrated significant potential and improved the efficiency and effectiveness of self-attention, several challenges remain unsolved. This section highlights the key limitations that still restrict the scalability and general applicability of VTs across diverse domains in computer vision along with the discussion of future research directions.

The performance of local windows is worse than that of the neighboring window because local windows may not be enough to facilitate the exchange of information between each other. The drawback of the neighboring version is low efficiency and it is difficult to calculate queries and keys in sliding neighboring areas. Techniques like sliding chunks, unfolding operations, or custom CUDA implementations, which are resource-intensive, either in terms of memory or time are required. Understanding how to use comparative metrics other than the dot-product while still maintaining good performance is a significant challenge (Zhou et al., 2023).

9.1. High computational complexity and memory consumption

Most of the models developed in recent years aim to achieve efficiency while maintaining accuracy (Sandler et al., 2018), however, the quadratic complexity of self-attention with respect to the number of patches still poses a significant challenge. This is accentuated when working with high-resolution images or long sequences. Several state-of-the-art architectures have introduced promising solutions. For instance, Longformer (Zhang et al., 2021b) and BiXT (Hiller et al.,

Table 6
Comparative analysis of Vision Transformer models grouped by parameter size, showing parameter count (M), GFLOPs, and Top-1% accuracy on ImageNet-1K (224 × 224).

Size	Method	Param. (M)	GFLOPs	Acc. (%)
Extra-tiny	CoaT (Xu et al., 2021a)	10	6.8	81.0
	LeViT (Graham et al., 2021)	10.9	0.7	80.0
	EffiformerV2 (Li et al., 2023e)	6.1	1.3	79.7
	Castling-ViT (You et al., 2023)	10.5	0.98	79.6
	RVT (Mao et al., 2022)	10.9	1.3	79.2
	CMT (Guo et al., 2022a)	9.5	0.6	79.1
	EMO (Zhang et al., 2023)	6.1	0.9	79.0
	MixFormer (Chen et al., 2022b)	8.0	0.7	78.9
	LightViT (Huang et al., 2022)	9.4	0.7	78.7
	HRFormer (Yuan et al., 2021d)	8.0	1.8	78.5
Tiny	Pale-ViT (Wu et al., 2022a)	22	4.2	84.2
	Inception (Si et al., 2022)	20	4.8	83.4
	T2T-ViT (Yuan et al., 2021c)	21.5	17.1	83.3
	PaCa-ViT (Grainger et al., 2023)	21.1	5.4	83.2
	FLatten (Han et al., 2023)	21	4.3	83.1
	Shunted-SA (Ren et al., 2022)	22.5	4.9	82.9
	VSA (Zhang et al., 2022d)	20	5.6	82.7
	CoaT (Xu et al., 2021a)	22	12.6	82.1
	EffiformerV2 (Li et al., 2023e)	12.6	1.25	82.0
	CoaT-Lite (Xu et al., 2021a)	20	4.0	81.9
Small	Dual-ViT (Yao et al., 2023)	42.6	9.3	85.2
	UniFormer (Li et al., 2023c)	50	8.3	85.1
	Pale-ViT (Wu et al., 2022a)	48	9.0	85.0
	Wave-ViT (Yao et al., 2022)	34	7.2	84.8
	Inception (Si et al., 2022)	48	9.4	84.6
	Castling-ViT (You et al., 2023)	34.7	7.0	84.6
	CMT (Guo et al., 2022a)	45.7	9.3	84.5
	CoaT (Xu et al., 2021a)	45	28.7	84.5
	BiFormer (Zhu et al., 2023)	26	4.5	84.3
	MiniViT (Zhang et al., 2022c)	46	31.4	84.3
Base	Pale-ViT (Wu et al., 2022a)	85	15.6	85.8
	Dual-ViT (Yao et al., 2023)	73	18	85.7
	VOLO (Yuan et al., 2022)	86	20.6	85.4
	BiFormer (Zhu et al., 2023)	58	9.8	85.4
	VOLO (Yuan et al., 2022)	59	14.1	85.2
	Castling-ViT (You et al., 2023)	51.9	9.8	85.0
	Refiner (Zhou et al., 2021b)	81	19.1	84.9
	Inception (Si et al., 2022)	87	14.0	84.8
	CMT (Guo et al., 2022a)	74.7	19.5	84.8
	Slide (Pan et al., 2023)	78	15	84.7
Large	VOLO (Yuan et al., 2022)	193	43.8	85.7
	UniFormer (Li et al., 2023c)	100	12.6	85.6
	LV-ViT (Jiang et al., 2021b)	150	59.0	85.3
	MaxViT (Tu et al., 2022)	212	43.9	85.2
	GCViT (Hatamizadeh et al., 2023)	90	14.8	85.0
	DGT (Liu et al., 2022)	90	16.4	85.0
	XCiT (Ali et al., 2021)	189	36.1	84.9
	MaxViT (Tu et al., 2022)	120	23.4	84.9
	DeepViT (Zhou et al., 2021a)	185.9	39.0	84.7
	Scalable-ViT (Yang et al., 2022b)	104	14.7	84.4

2024) reduced complexity while handling longer sequences. While there is progress but, issues of limited input scalability and hardware-dependency still remain. These efforts often involve trade-offs between complexity and accuracy. As such, the fundamental tension between rich global context modeling and low-cost computation is still a balancing act, especially for deployment on memory-limited devices or performance-critical scenarios. Potential solutions are needed to handle such data on resource-limited systems with improved accuracy.

A deeper issue in this context is the handling of patches. Often, all patches are processed with equal attention cost, regardless of semantic importance. Future work could target input-adaptive computation and modular architectures whose capacity scales with task complexity instead of pixel count. For a simple image, the model might use fewer layers or attention heads, whereas a complex scene could trigger a deeper computational path. The core challenge is how to form these decisions, and what these decisions are based on.

9.2. Data efficiency and dependence on large-scale datasets

VTs typically require massive amounts of training data to achieve good performance. This reliance limits their application in domains where such extensive labeled data is not available. Even with large datasets, VTs may not utilize data as efficiently as CNNs. Studies have indicated that VTs can suffer from over-smoothing issues where feature representations across distinct classes become similar, affecting model performance (Zhou et al., 2023). This is partly due to the Transformer's lack of inherent inductive bias, which can make it harder for VTs to generalize well to domain-specific tasks without large amounts of domain-specific data or using limited datasets. Dynamic token selection, similar to Rao et al. (2021), can be used to prioritize informative patches to reduce redundancy. More improved models like Meng et al. (2022) and Yin et al. (2022) can be developed to utilize the adaptive tokens strategy. Hybrid self-supervised frameworks combining VTs with

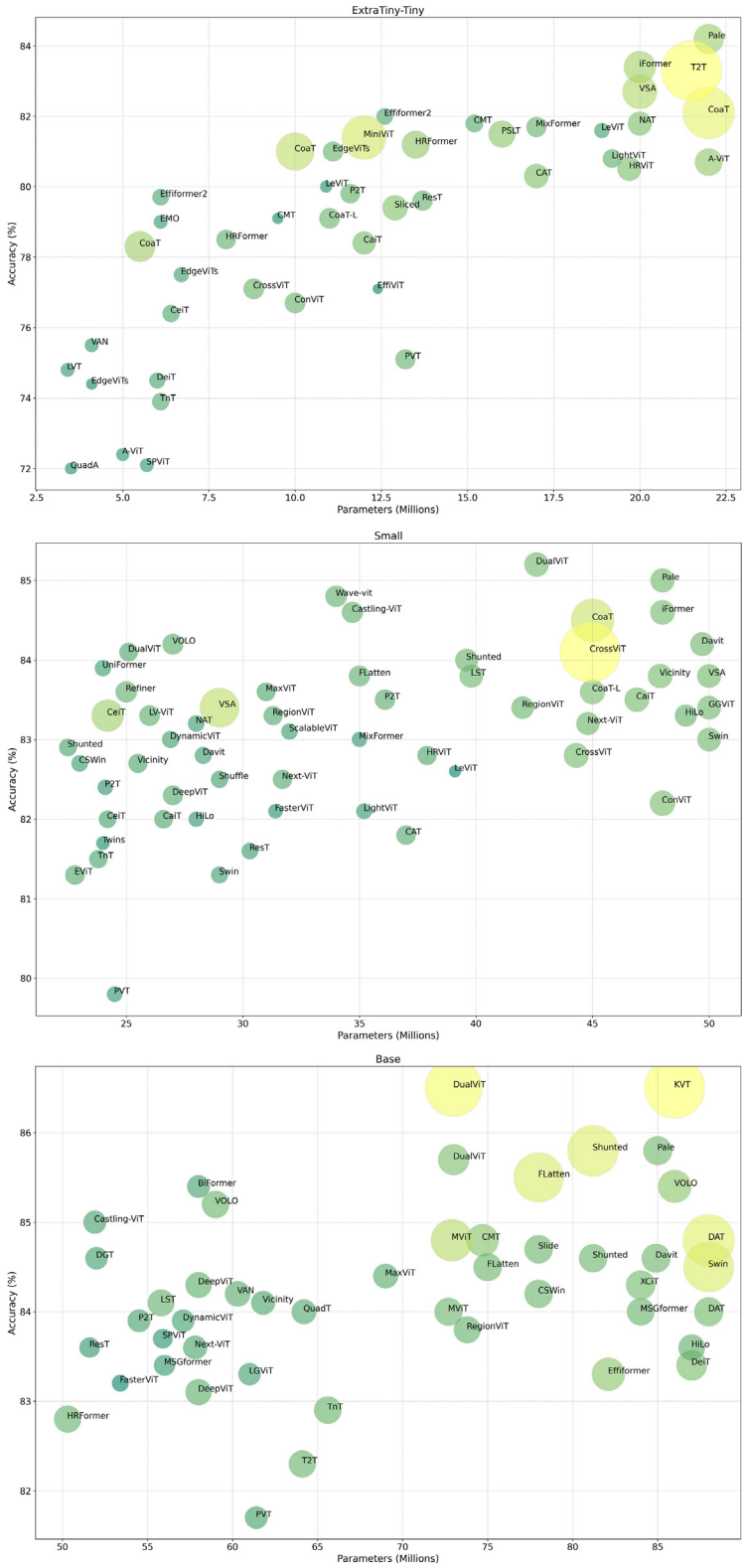


Fig. 7. Vision Transformer (VTs) Models: ImageNet top-1% Accuracy vs Parameters(M) vs Operations(GFLOPs). Bubble size represents computational cost (GFLOPs). The graphs are categorized into parameter size groups: ExtraTiny to Tiny, Small, Base and Large.

generative models i.e. Dunlap et al. (2023) to synthesize domain-specific data, could boost performance on small datasets where data are scarce.

VTs fundamental component is the self-attention mechanism and approaches often adapt components from earlier backbone architectures such as Wang et al. (2021). This adaptation enhances performance but also inherits many limitations of the original methods.

This reliance on inherited designs also limits exploration of alternative attention mechanisms that could potentially overcome inherited challenges. Therefore, there remains substantial room for research into novel attention techniques which could lead to more efficient, robust architectures.

9.3. Architectural challenges

Multi-head attention is designed to learn diverse feature representations but studies show that attention heads often learn redundant or overlapping representations. Techniques like head pruning (Voita et al., 2019), orthogonality constraints (Zhang et al., 2021a), and disentangled attention (He et al., 2021) have been proposed to reduce this redundancy, but they often involve trade-offs in performance. So far, no method guarantees efficient and diverse head utilization without loss in accuracy. Exploring dynamic head allocation could offer promising directions.

It is hard for local windows to exchange information between each other. It was solved by other window techniques, such as neighboring windows (Hassani et al., 2023) but these have their own limitations. Computing queries and keys across neighboring areas is difficult and inefficient. It needs techniques like sliding chunks, unfolding operations, or custom CUDA implementations, which are resource-intensive and complex. More efficient and scalable mechanisms for cross-window communication remain an open challenge.

As the depth of VTs increases, feature representations stop evolving and limit the model's ability to capture diverse concepts, leading to attention collapse. There are some solutions, such as Zhou et al. (2021a), Touvron et al. (2021b) however, scaling VTs in a stable manner is still a challenging task. In parallel, state space models such as Mamba (Gu and Dao, 2024) have emerged as an alternative sequence modeling paradigm. Rather than replacing Transformers, recent work explores hybrid designs that combine selective state space modules with attention layers, uniting recurrence-driven scalability with content-adaptive attention. Although Vision-Mamba models (Liu et al., 2024b) provide linear-time sequence modeling and improved scalability, several challenges remain. They do not provide fully query-specific pairwise interactions like self-attention, which can limit fine-grained relational reasoning in dense prediction and compositional tasks. As a result, hybrid attention and state space architectures (Hatamizadeh and Kautz, 2025) have become a central direction, focusing on complementary integration rather than replacement. Adapting fundamentally one-dimensional recurrent formulations to structured two-dimensional data also remains challenging. Multi-directional scanning, windowed state space modules, and hierarchical state propagation improve spatial inductive bias, yet a standardized two-dimensional state space design paradigm comparable to convolutional or Transformer backbones has not fully emerged.

9.4. Generalization and robustness

Although self-attention can capture long-range dependencies, VTs still struggle with robustness to adversarial attacks, occlusions, lighting conditions or other real-world image degradations. Studying how self-attention mechanisms respond to adversarial attacks and finding ways to make them more robust is also essential. New training methods can be designed that resist manipulation while maintaining performance. Due to the lack of inductive bias, self-attention fails to capture visual patterns when data is scarce (Xu et al., 2021b). Future work could adapt inductive bias into self-attention to enhance generalization under such conditions.

9.5. Other challenges

Most of the self-attention models compute query and key independently, however, some studies (Tay et al., 2021) that integrate query and key representations have achieved competitive performance. This suggests that the independent representation of these components may not be essential, leaving the question open for further research (Niu et al., 2021).

Despite the observed progress, achieving generalizable performance across diverse datasets and tasks within image classification remains a challenge. It is still challenging to design models that handle different image resolutions, object scales or domain shifts along with handling tasks even within image classification. State-of-the-art models still rely on task or dataset-specific design choices such as data augmentation, architectural tuning, or pretraining on large-scale datasets to perform well. Developing a more task-agnostic and data-efficient learning framework for image classification remains an open challenge and a promising direction for future research, enabling broader generalization without heavy reliance on domain-specific models.

It has been observed that attention mechanisms require a preliminary analysis to understand the best training procedures and settings. Therefore, investigating new optimization methods such as You et al. (2019) specifically for attention models is likely to be worthwhile.

The development of efficient and effective models that are inspired by the human visual system remains a relatively unexplored area of research, except for a few examples such as Kockwelp et al. (2025), Eberle et al. (2022). This field holds great potential, as understanding how humans process visual information can lead to the creation of more adaptive and attentive vision systems.

10. Conclusion & discussion

It has become more common to use vision Transformers (VTs) instead of CNNs because of ViT's robust performance in almost every vision task. Attention mechanisms play a crucial role in enhancing the efficiency and accuracy of these VTs. This survey provides a comprehensive review of various self-attention approaches employed in VTs. The survey begins with the introduction of the vision Transformer (ViT), attention mechanism, self-attention and multi-head attention. Then it explores the advanced vision Transformers (VTs), categorized by different attention techniques. The first criterion discussed is self-attention fusion with convolution attention where models use convolution operations along with self-attention to obtain local information more effectively than others. It uses the strength of both local and global contexts and works best for constrained devices and real-time applications. Next, the hierarchical attention is covered in which down-sampling or upsampling techniques are used to make the input of different scales allowing the extraction of diverse types of information. It helps to build representations from fine to coarse, mimicking CNN hierarchies, which is essential for dense prediction tasks. Cross-attention is also discussed where information between different input sequences or branches is exchanged which helps to improve the accuracy. Similarly, attention fusion is reviewed where different attention mechanisms exchange information and capture diverse feature relationships which is helpful for complex vision tasks. Further, the survey discusses window-based attention which applies different types of windows on top of patches to capture more localized information. Axial attention also tries to gather local information by using horizontal, vertical or diagonal windows. Both of these mechanisms gather local information along with local context effectively and attempt to lower the computational power while maintaining accuracy. Token manipulation in attention is also addressed where various methods try to modify tokens interaction with attention by focusing on relative parts. This helps reduce the memory consumption of the model. Finally, the survey reviews techniques aimed at modifying the original formula of self-attention to make it more efficient. It uses various techniques to

calculate attention scores to improve the feature representations. It also obtains targeting efficiency and better positional encoding.

Overall, tokens manipulation, windows, and axial attention achieve better efficiency using less computational power and memory consumption. Hierarchical, cross and fused attention are capable of achieving good performance. We also discovered that convolution operations and token manipulation are the top categories used in most of the models.

This survey provides a comprehensive view of the previous attention mechanisms in VTs developed since the original ViT paper, highlighting their major contributions and providing a unique view according to the taxonomy.

The prevailing trend indicates that the most successful Vision Transformer designs do not replace softmax attention but constrain, organize, and complement it with structural inductive bias and multi-scale hierarchy. The winning recipe is not a single innovation but a combination of hierarchical structure, localized attention, controlled global interaction, and hardware-aware implementation.

Throughout this survey, we have observed a clear evolution in Vision Transformer design. Early efforts primarily targeted the reduction of computational cost while preserving accuracy, often through structural locality constraints and improved local–global interaction mechanisms. More recent developments extend beyond efficiency, emphasizing task-specific adaptation, foundation-scale pretraining, multimodal and generative modeling, and the emergence of post-ViT paradigms such as pure and hybrid state space models. At the same time, increasing attention to interpretability, robustness, and sustainability reflects the growing maturity of the field. Collectively, these trends indicate that future vision architectures will likely integrate attention, recurrence, multi-scale hierarchy, and explainability-aware design within unified and scalable frameworks. We hope this survey provides a principled foundation for understanding these developments and supports the design of more accurate, efficient, and transparent vision systems that are environmentally friendly.

CRediT authorship contribution statement

Ahmad Waseem: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Pietro Ruiu:** Writing – review & editing, Methodology, Conceptualization. **Seth Nixon:** Writing – review & editing, Methodology. **Andrea Lagorio:** Writing – review & editing, Methodology, Conceptualization. **Massimo Tistarelli:** Supervision, Project administration, Funding acquisition.

Declarations of funding

This work has been developed within the framework of the following projects:

1. “Visual Attention for New-generation of Vision Transformers (VisAViT)” - CUP J83C24000230007, funded under the Public Call for the selection of project proposals, aimed at granting funds for activities consistent with the “Future Artificial Intelligence Research - FAIR” Project - code PE00000013 - CUP J53C22003010006 – under the National Recovery and Resilience Plan (PNRR) Mission 4 Component 2 Investment Line 1.3 funded by the European Union – NextGenerationEU.
2. e.INS- Ecosystem of Innovation for Next Generation Sardinia (cod. ECS 00000038) funded by the Italian Ministry for Research and Education (MUR) under the National Recovery and Resilience Plan (NRRP) - MISSION 4 COMPONENT 2, “From research to business” INVESTMENT 1.5, “Creation and strengthening of Ecosystems of innovation” and construction of “Territorial R&D Leaders”;

A.L. and M.T. are funded by 1. S.N., A.L. and M.T. are funded by 2. A.W. and P.R. are not funded by the above projects.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to thank all of our colleagues involved in this research for their time, engagement, and support.

Data availability

We do not analyze or generate any datasets, as our work is a review of publicly available research. The original versions of these data can be found in the references below.

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